Cyber AI Resistant Secuity using KERI+ACDC+Survivability Engineering: Tunable Fault Tolerant System

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Preliminaries

Computational infeasiblity of strong crypto is not vulnerable to a brute-force attack (Cyber AI or not)

Full granular cryptographic agility means pre-migrated to post-quantum capable

The principal weakness of all cryptographic security systems are side-channel attacks

KERI+ACDC is a signed everything, no-shared-secret security system.

Its principal weakness to side-channel attacks is the management of private keys.

KERI is a tunable fault-tolerant DKMI based on a complementary set of threshold structures that can make system failure practically infeasible.

Preliminaries ...

Fault-tolerant system: Continues operating despite faults

A fault occurs when some part of the system

A failure (system failure) occurs when the system as a whole is no longer able to complete its mission.

No single point of failure

No common-mode failure sources

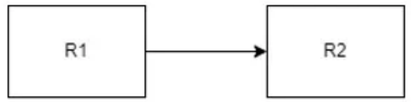
Repair faults as they occur so a cascade of faults does not lead to failure

Never fail to detect a fault

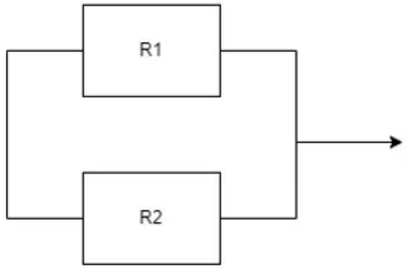
Surviveability Engineering

Adaptation of Reliability Engineering to highly dynamic, hazardous threat environments.

Combines reliability engineering techniques with critical infrastructure survivable risk mitigation design techniques.



$R_s = R_1 \cdot R_2$

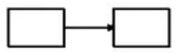
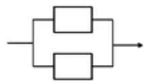


$R_s = 1 - (1 - R_1) \cdot (1 - R_2)$

Criticality = (likelihood of failure) * (consequence of failure)

Rate and rank infrastructure threat mitigation on criticality

Establish a risk profile based on threat environment

Formula	*Basic Equation*	*System Reliability*	*MTTF*
	$R_s = R_1 \cdot R_2$	$R_s(t) = e^{-\lambda_1 t} \cdot e^{-\lambda_2 t}$	$MTTF = \frac{1}{\lambda_1 + \lambda_2}$
	$R_s = 1 - (1 - R_1) \cdot (1 - R_2)$	$R_s(t) = 1 - (1 - e^{-\lambda_1 t}) \cdot (1 - e^{-\lambda_2 t})$	$MTTF = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{(\lambda_1 + \lambda_2)}$

Surviveability

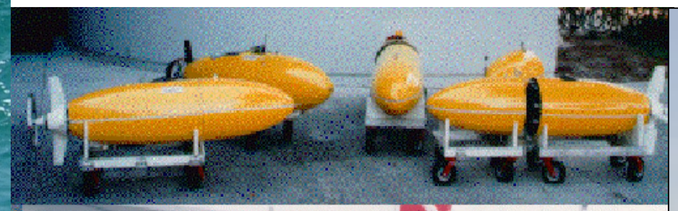
Susceptibility = likelihood of attack

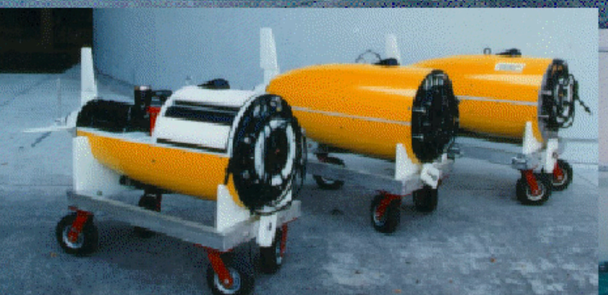
Vulnerability = likelihood of a damage consequence level given an attack

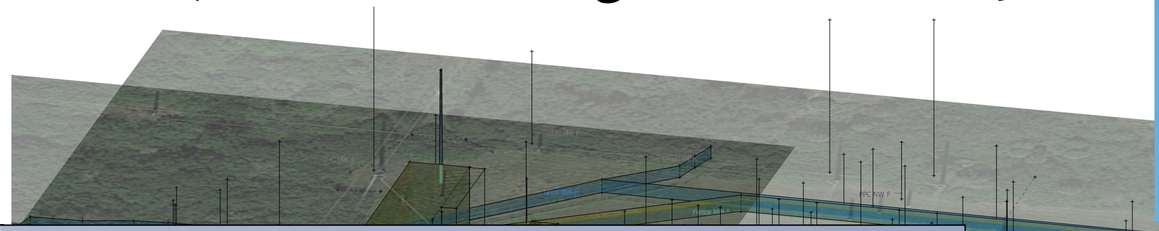
Recoverability = likelihood of damage repair in time to continue operations

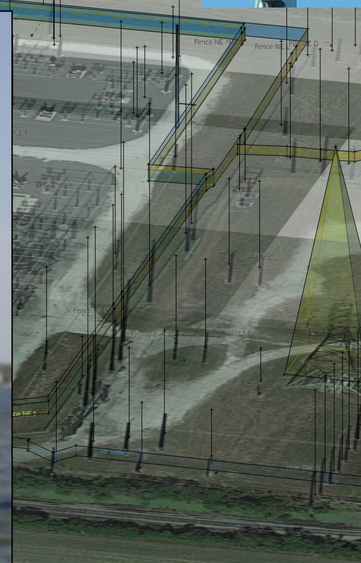
*Liklihood of Failure = Susceptibilty * Vulnerability * (1 - Recoverability)*

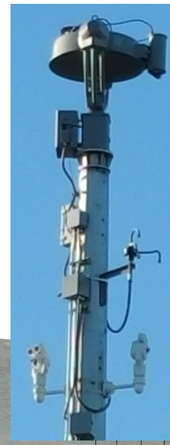
Insight: The most cost-effective survivability comes from increasing recoverability











Surviveability

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Threshold Structures

A threshold structure multiplies the number of attack surfaces that must be simultaneously compromised for a successful attack

A fault-tolerant system detects and recovers from a partially successful attack, thereby reverting the attack progress

The result is a Perpetually Secure System

- *Recoverable fault is a side-channel attack on signing keys.*
- *Loss of control over the identifier is a system failure*

Complementary Threshold Structures

KELs +

Signing Keys and prerotated rotation keys (friction reduction and recovery)

Multi-sig signing keys

Multi-sig rotation keys

Hierarchy of Delegated AIDs (non-common mode separation and recovery)

Witness pools (fault detection and Fault resilience)

Watcher network

TELs + ACDCs (fault detection

Faults and Fault Recovery

A successful attack must start with private key signing infrastructure compromise.

Pre-rotation enables recovery from private key signing infrastructure compromise

AID delegation enables recovery of private key rotation infrastructure compromise

System failure when non-delegated AID rotation keys are compromised

System failure when delegation hierarchy of rotation keys are all compromised

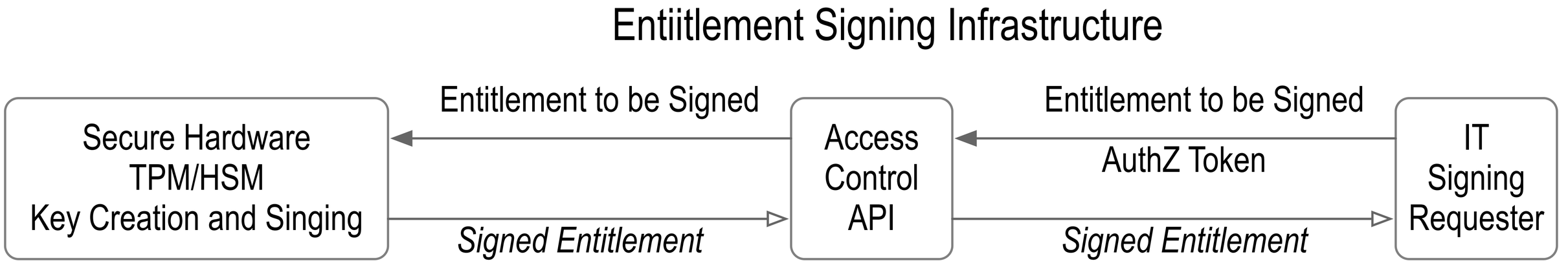
Witness pool enables detection of key compromise used to attack key state

Witness pool enables detection of key compromise used to attack bound issuances

Governance as a fault:

Split infrastructure avoids common-mode governance faults

Conventional Enterprise Signing Infrastructure



Best practice is to use secure signing hardware (TPM, HSM, Secure Enclave).

The private key never leaves the hardware. The public key is broadcast to verify the signature.

Role-based or capability-based access control grants AuthZ permissions/tokens to employees who submit entitlements to secure hardware for signing.

The weakest vulnerability is the AuthZ infrastructure (typically OAuth/OIDC/SSO based)

, not the private key signing hardware.

Systems that do not use secure signing hardware are also vulnerable to direct private key compromise

Tunable Parameters

Establishment Only KEL (all private keys are one-time use)

Signing Key(s): Single-Sig versus Multi-Sig

Multi-sig threshold, fractionally weighted versus simple threshold, nested

Rotation Key(s): Single-Sig versus Multi-Sig

Multi-sig threshold, fractionally weighted versus simple threshold, nested

Custodial Split of Signing Keys vs Rotation Keys

Witness pool threshold and infrastructure-dependent fault mitigation

Delegated AIDs:

Multi-valent fault environment

Horizontally scaled infrastructure

Watcher network configuration

Fault Tolerant Fault Detection

Good fault-tolerant design starts by designing the system so that faults are always exposed.

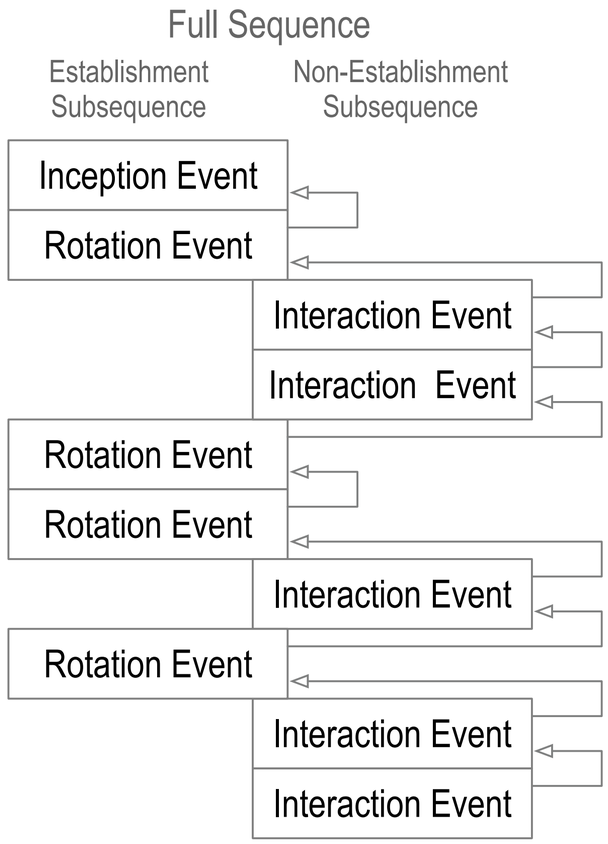
Make faults detectable. Don’t hide them.

Better the fault you know about but can’t fix just yet than the fault you don’t know about, even if you don’t need to fix it just yet.

The faults you don’t know about will always compound until eventually you have an unanticipated catastrophic system failure.

The sentence above accurately describes just about every security breach.

Inconsistency and Duplicity



inconsistency: lacking agreement, as two or more things in relation to each other

duplicity: acting in two different ways to different people concerning the same matter

Internal vs. External Inconsistency

Internally inconsistent log = not verifiable.

Log verification from a self-certifying cryptographic root-of-trust protects against internal inconsistency.

Externally inconsistent logs. Two different logs for the same identifier, both verifiable = duplicity.

Duplicity detection protects against external inconsistency.

KERI provides duplicity evident DKMI

Duplicity Detection

KERI's aha moment was realizing that a compromised controller and a malicious controller can be forced to exhibit the same fault: *duplicity*.

External inconsistency == duplicity.

One can trust as long as the non-repudiable signatures verify and there is no evidence of duplicity.

One stops trusting at the first sign of duplicity.

One does not resume trusting unless that duplicity has been reconciled.

KERI forces all faults to exhibit as duplicity.

Given that, fault detection becomes duplicity detection. This simplifies fault detection.

Given a fault-tolerant duplicity detection mechanism, all other faults are detectable. KERI makes all computation duplicity-evident, so the detection mechanism is endemically fault tolerant.

KERI makes the cost of compromising duplicity detection higher than the return gained from doing so.

Signed, Hash Chained, Everything

The way KERI forces all faults to exhibit, i.e., be detectable as duplicity, is to use only digital signatures, no shared secrets.

A digital signature requires the signer to know the private key.

When something is verifiably signed, either the signer acted maliciously when they signed it, or their signing infrastructure was compromised.

KERI forces duplicity-evident computation by using signed, hash-chained data structures everywhere that become provably duplicitous when they are not both internally and externally consistent.

All successful attacks can be forced to exhibit in a detectable way.

Tunable Security on Top of Duplicity Detection

KERI's security rests on a single fault type, with a fault-tolerant detection mechanism for that fault type.

Layered threshold structures make it exponentially harder to attack that fault detection mechanism cost-effectively.

Recovery mechanisms force the attacker to start over, effectively making the ROI of a successful attack negative.

Tunability lets you trade off friction, fault resilience, and fault tolerance for each risk profile.

Once one groks this, it becomes easier to understand the way of KERI.

AI Surveliance Problem

Continuous Identity

Behavioral Biometrics

Participatory Identity

Persona Based Authentication

Authentication via Statistical Correlation

AI Surveliance Problem …

The more pervasive the surveillance,
the higher the fidelity the model,
the more likely it will distinguish a real person from a fake

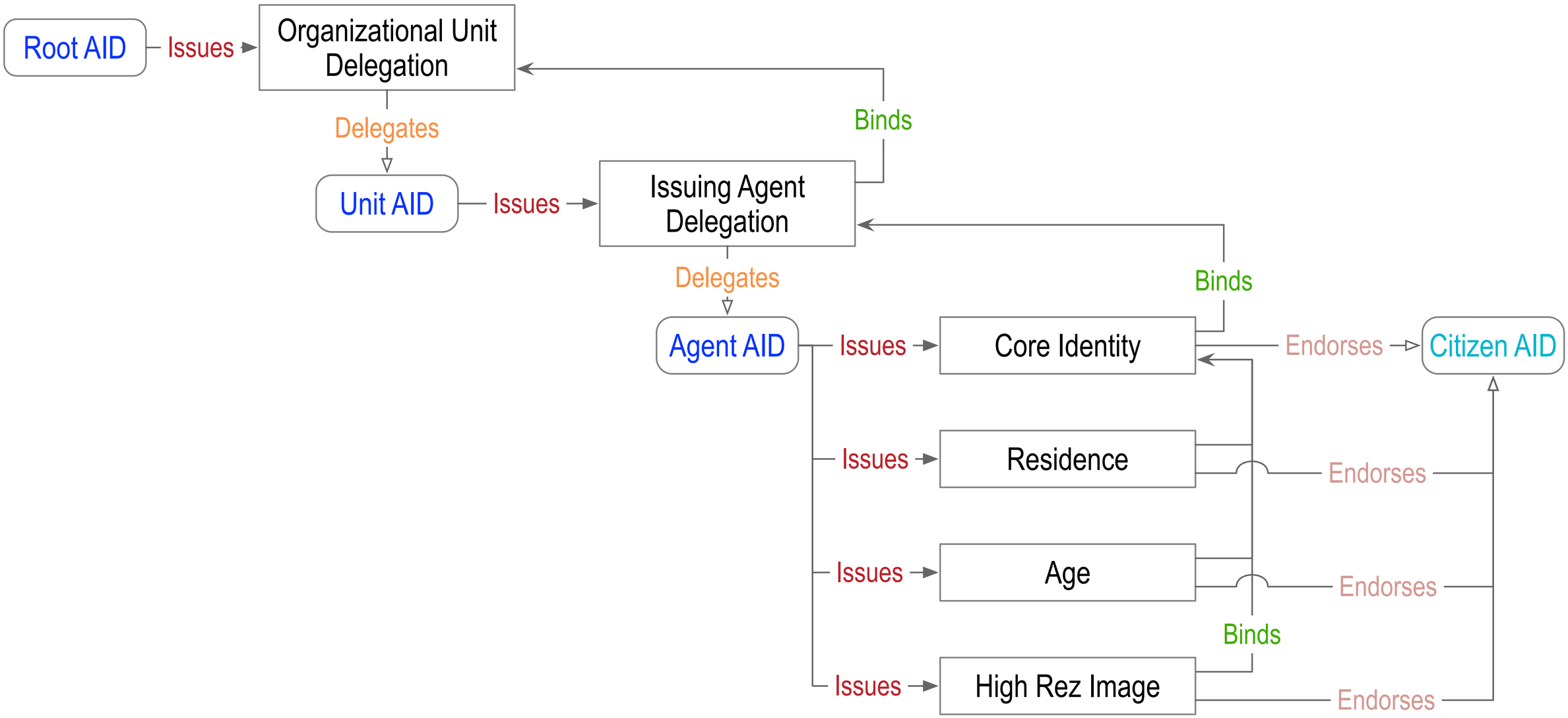
Arms race between external (unpermitted) surveillor and internal (permitted) surveillor
as both work by aggregation and correlation of increasingly pervasive observations of behavior.

Any breach of the internal surveillance data immediately destroys the protective delta between the internal & external surveillors.

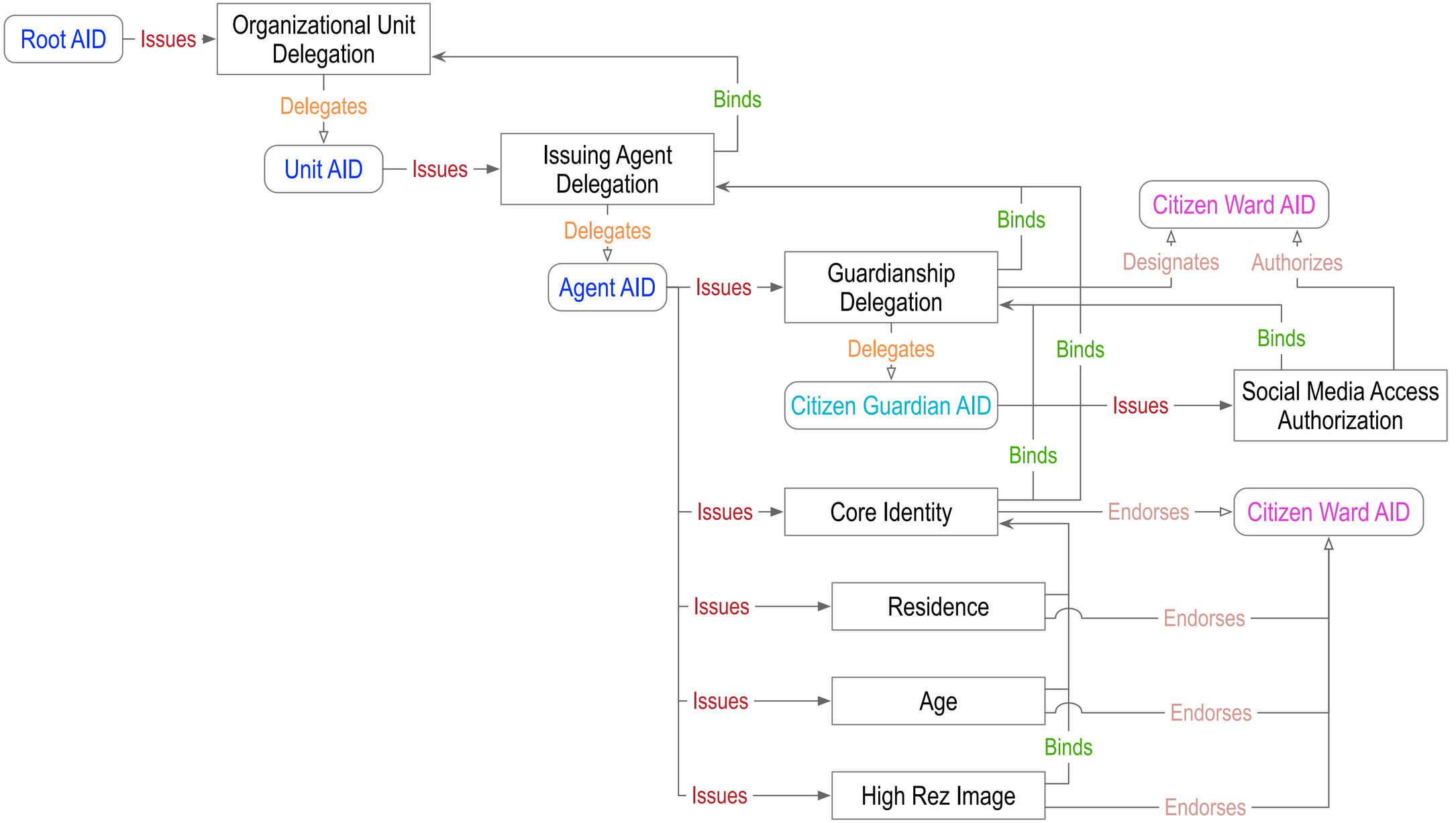
Solution CAN NOT, MUST NOT be ...

more survelliance

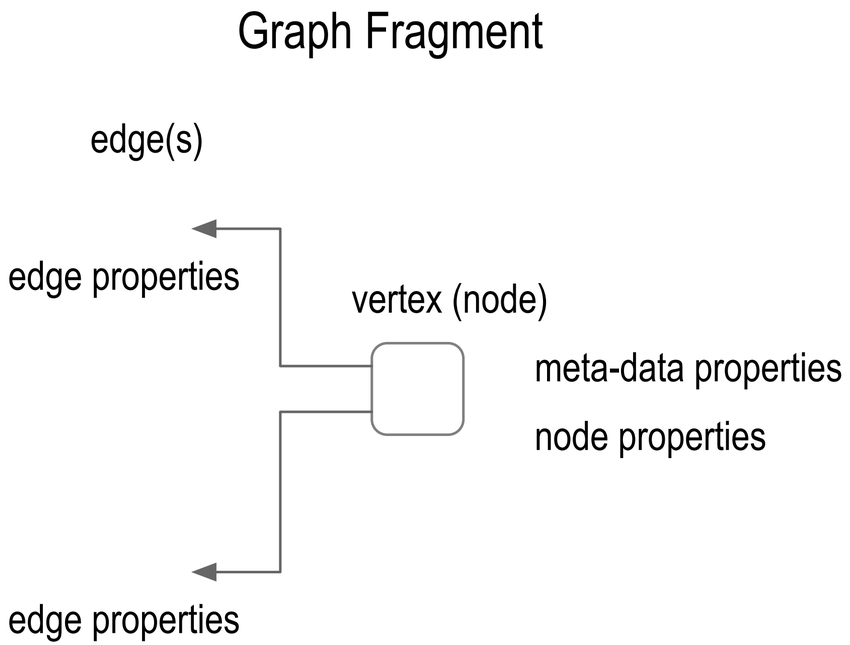
SEDI: Citizen Sourced State Endorsed

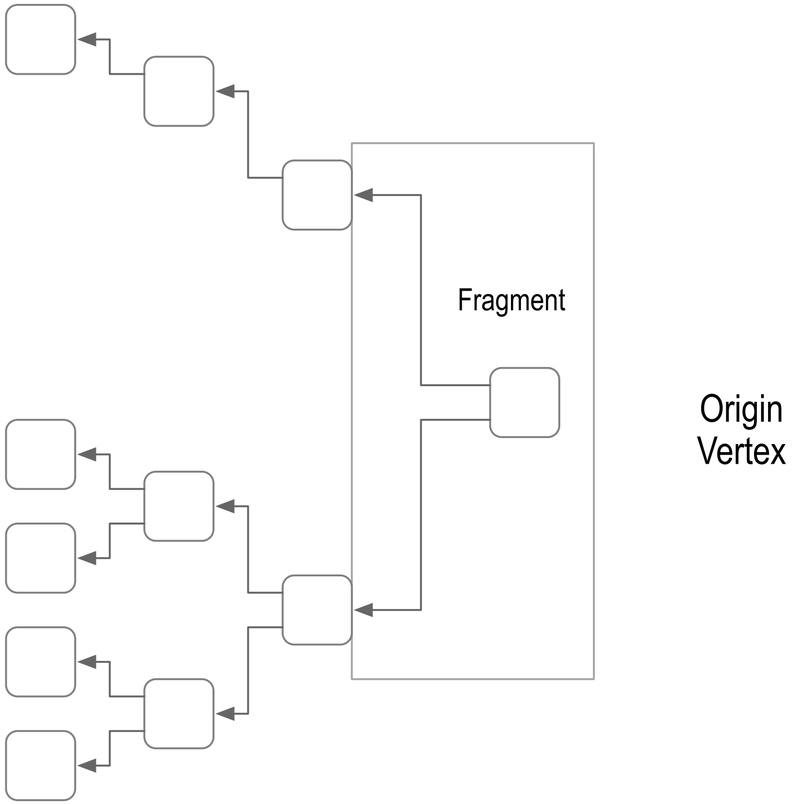


SEDI: Guardianship



Using ACDCs





Perpetually Verifiable Issuances

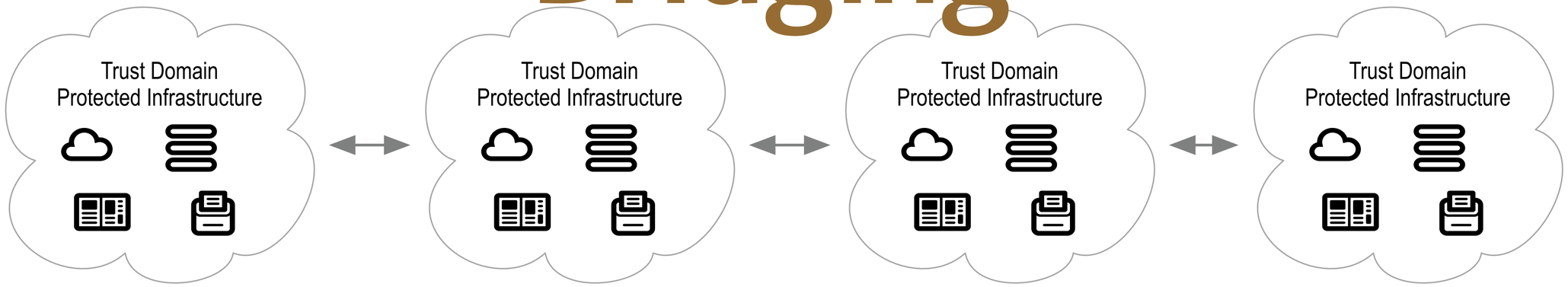
Each Fragment is Separably Authenticatable

Authority: (DAAART)

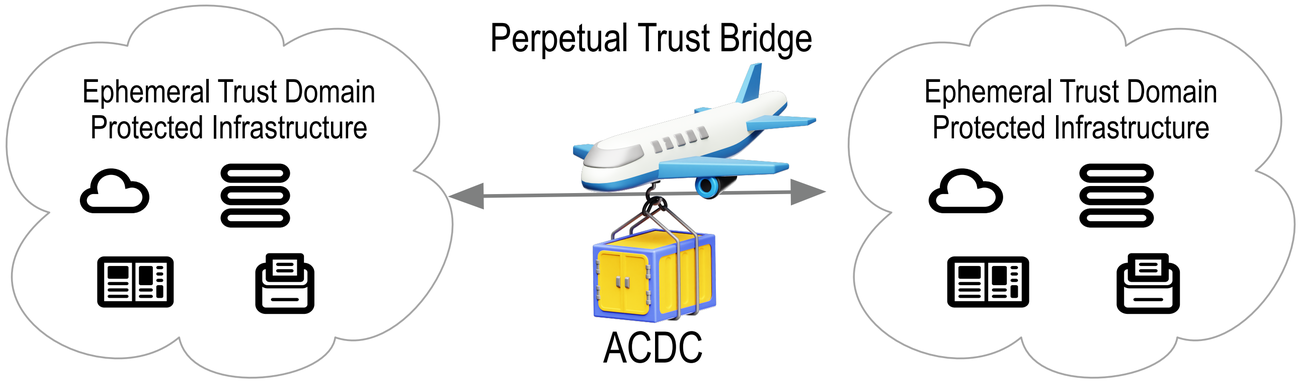
Delegable Attenuable Aggrable Auditable Revocable Transferable

High Fidelity Modeling of Real World Authority Structures

Bridging



Moving Trust Across Trust Domains



perpetual identity → perpetual issuances → lossless trust transfer (KERI Suite)
ephemeral identity → ephemeral issuances → lossy trust transfer (DNS/CA, OAuth/OIDC)
ACDC lossless conveyor of trust → bridging ephemeral trust domains
eliminates recursive privilege escalation attack vulnerabilities

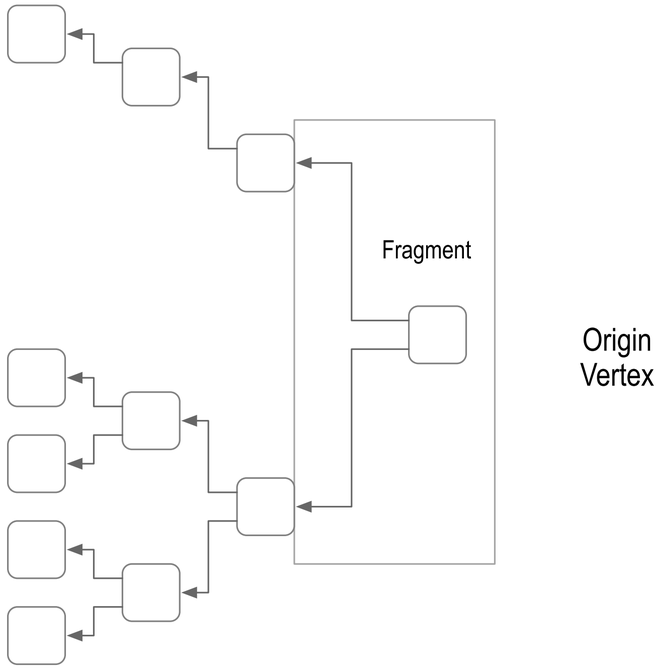
Agentic Control

Agent as a verifiable authorized delegate of a natural person or legal entity
Agent has its own AID
Realizes the original meaning of the word *agent*
ACDC Mechanisms are the same.
Attributes that indicate the type of agent. (AI vs. person)

Edge Verifiable Agentic Control (EVAC)

Agentic Authorization

Authorization to access a resource is granted by the resource's controller (authority)
Each authority (principal) is itself an edge with respect to other authorities' resources
Each authority can authorize access by any other edge to its resources via an ACDC
Each Issuer in an ACDC graph is potentially both a resource authority and an edge.
Mutually authenticated edge accessor and resource authority prevent impersonation attacks



Edge Verifiable Agent Control (EVAC)

Edge Verifiable Agent Control (EVAC)

ACDC

v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Issuance Registry

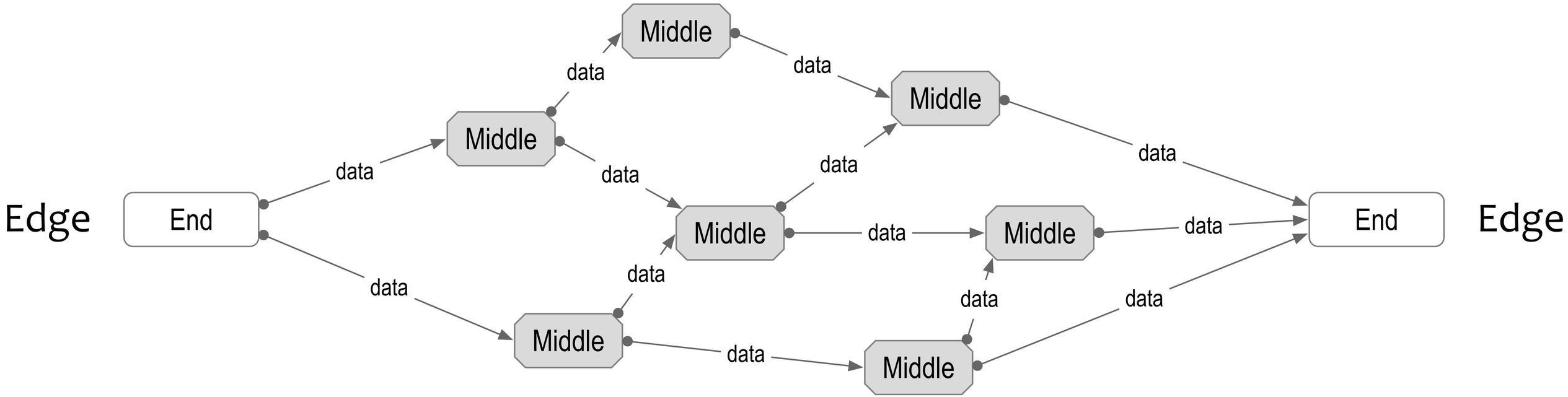
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Usage Registry

SAID	UUID	Issuee AID	Registry	Resource Map

route1	capabilities1
route2	capabilities2
…	…

End-to-End — Edge

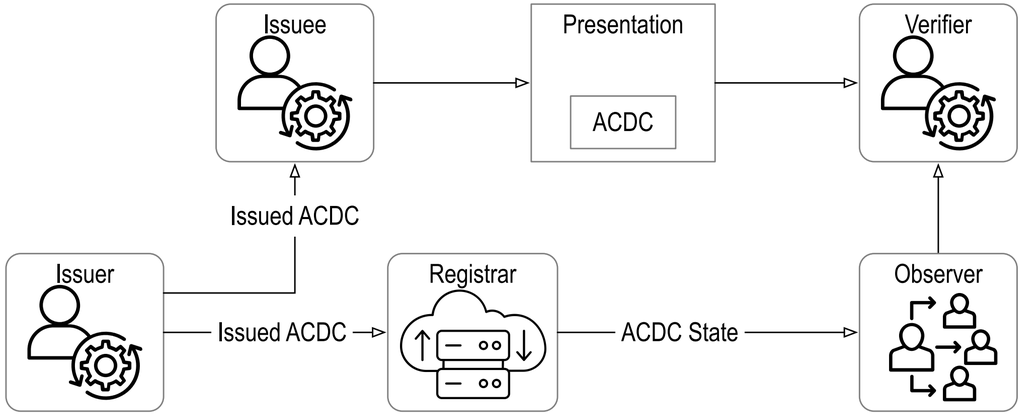


With KERI, each AID controller is its own identity provider

Keys-at-the-Edge (KATE) = each edge self-authenticates

No third-party IdPs, no relying parties, no trusted intermediaries

Issuance Registry of EVAC Authorization



ACDC

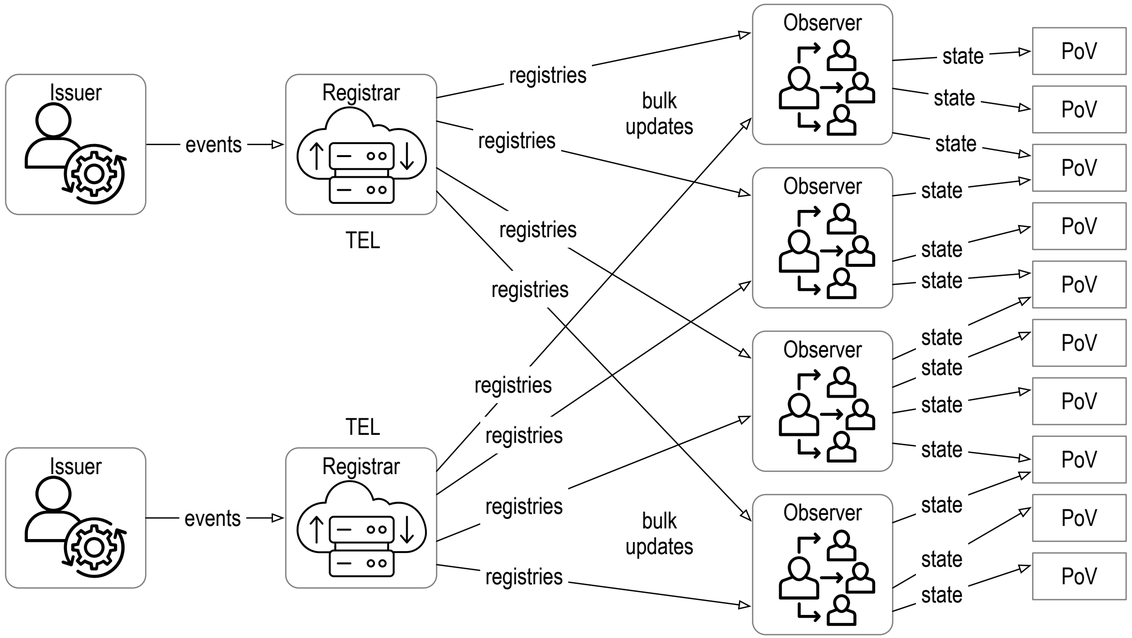
v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Registry Inception Event (rip)

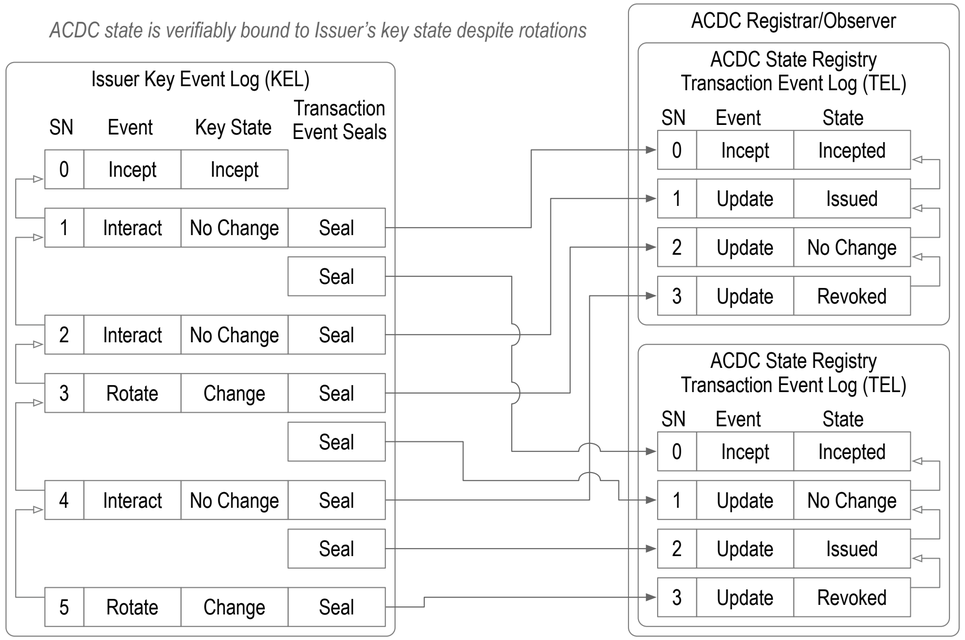
v	t	d	u	i	n	dt
Version	Type	SAID	UUID	Issuer AID	SN	DateTime

Registry Blinded Update Event (bup)

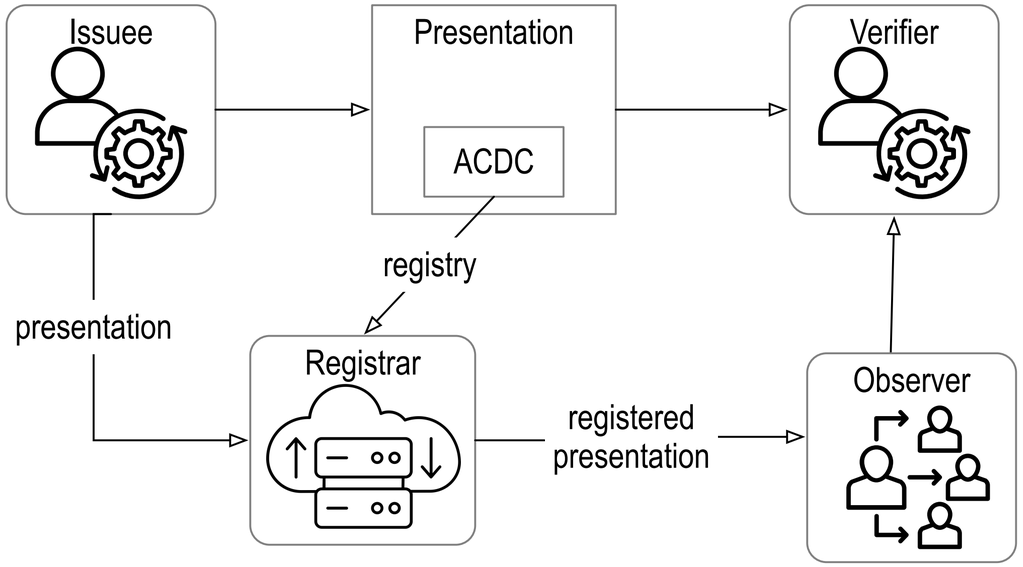
v	t	d	rd	n	p	dt	b
Version	Type	SAID	Registry	SN	Prior	DateTime	BLID



ACDC state is verifiably bound to Issuer's key state despite rotations



Usage Registry of EVAC Authorization



Issuance Registry

Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Usage Registry

SAID	UUID	Issuee AID	Registry	Resource Map

route1	capabilities1
route2	capabilities2
…	…

ACDC Issuee Presentation state is verifiably bound to Issuee's key state despite rotations

Issuee Presenter Key Event Log (KEL)

SN	Event	Key State	Transaction Event Seals
0	Incept	Incept	
1	Interact	No Change	Seal
2	Interact	No Change	Seal
3	Rotate	Change	Seal
4	Interact	No Change	Seal

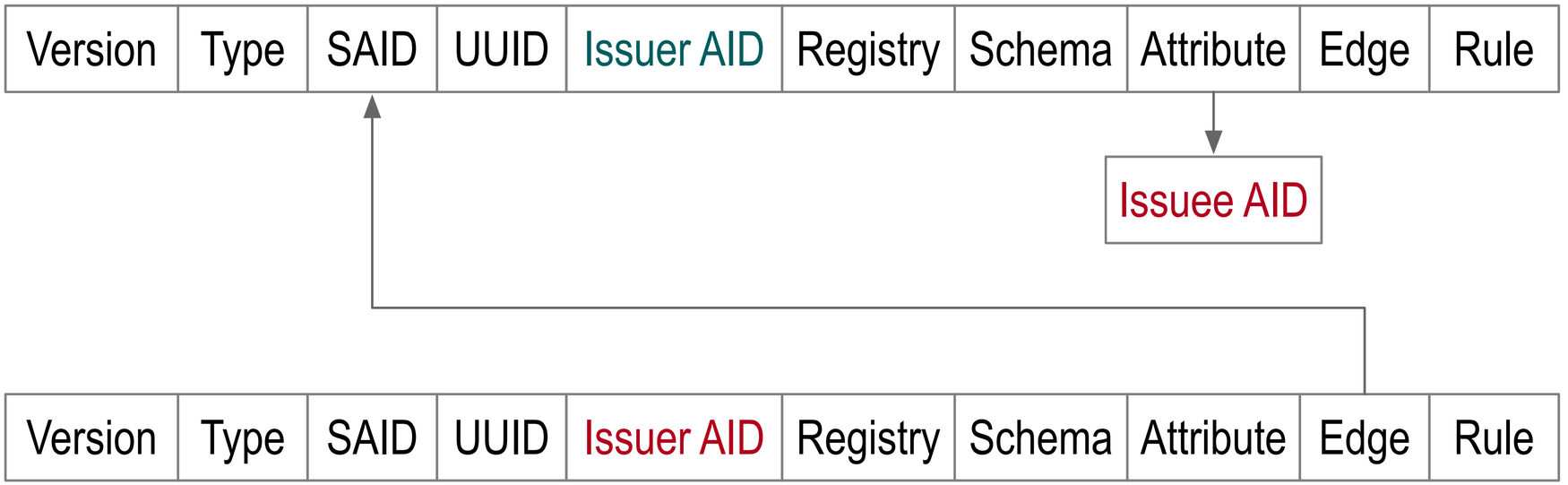
Issuee Presentation Registrar/Observer

Presentation Registry
Transaction Event Log (TEL)

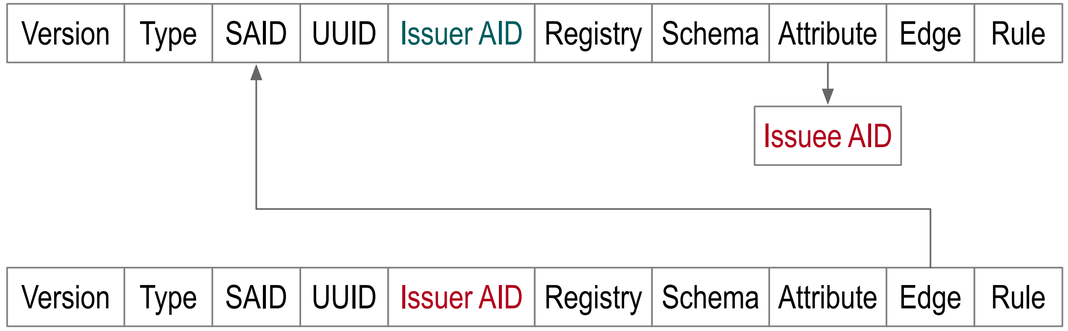
SN	Event	State
0	Incept	Incepted
1	Update	Presented
2	Update	No Change
3	Update	Presented

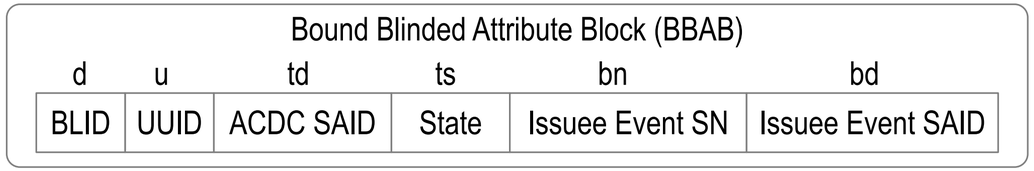
Reversion of Delegation

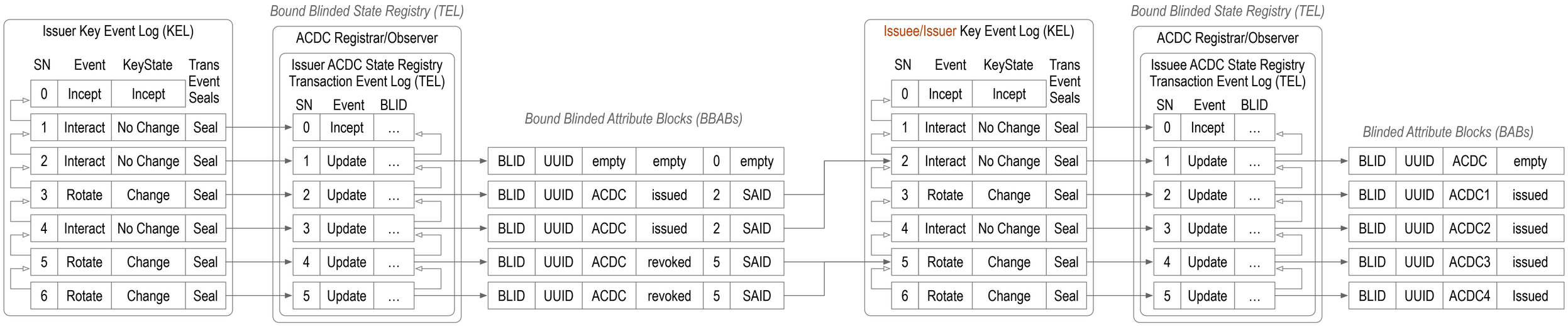
Revocable delegated authority problem:
When a delegatee's authority to issue authorizations is revoked, what is the status of any authorizations/entitlements issued before the authority was revoked?
How can a validator securely verify the status of prior valid issuances from a chain-of-authority when the chain is subsequently broken?



Bound Blinded ACDC Registry Recovation

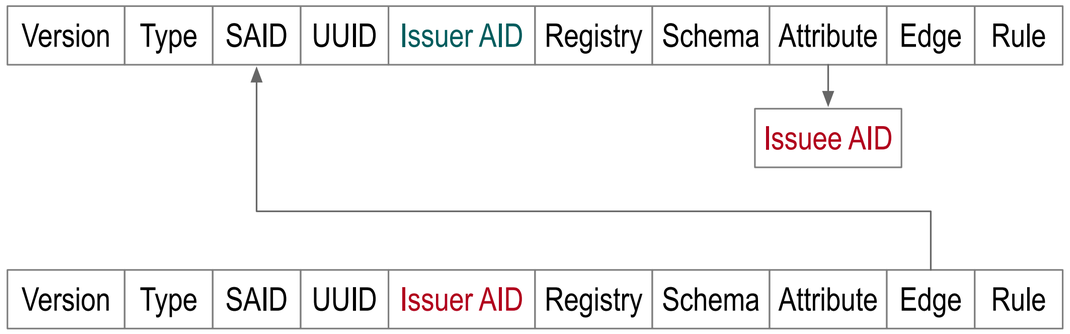


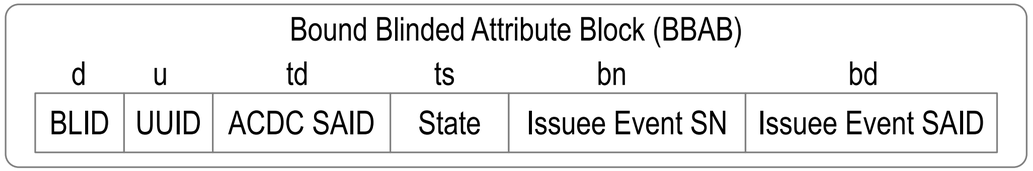


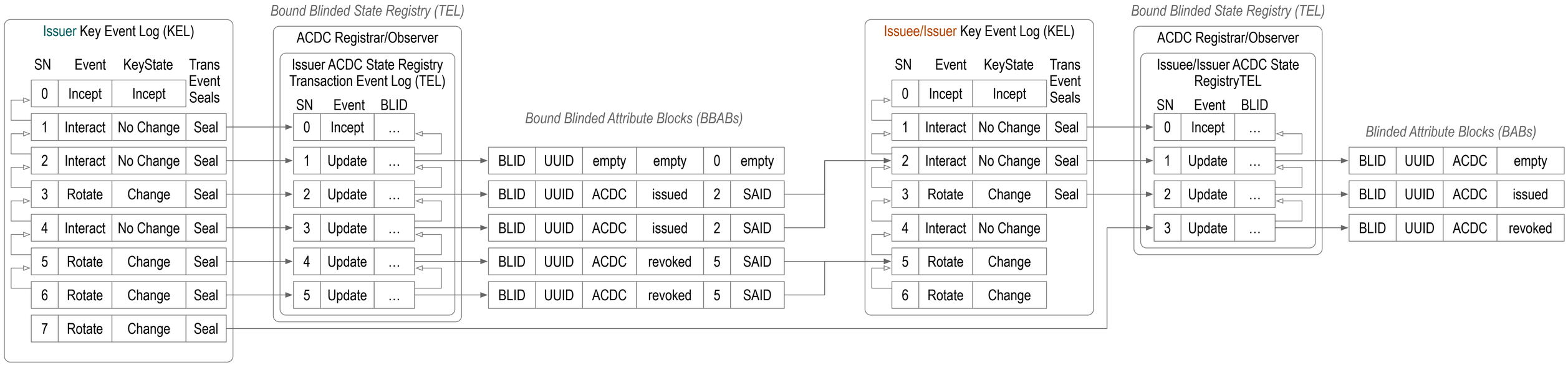


Breaking the chain-of-authority while preserving validity of prior valid issuances

Bound Blinded ACDC Registry Authority Reversion







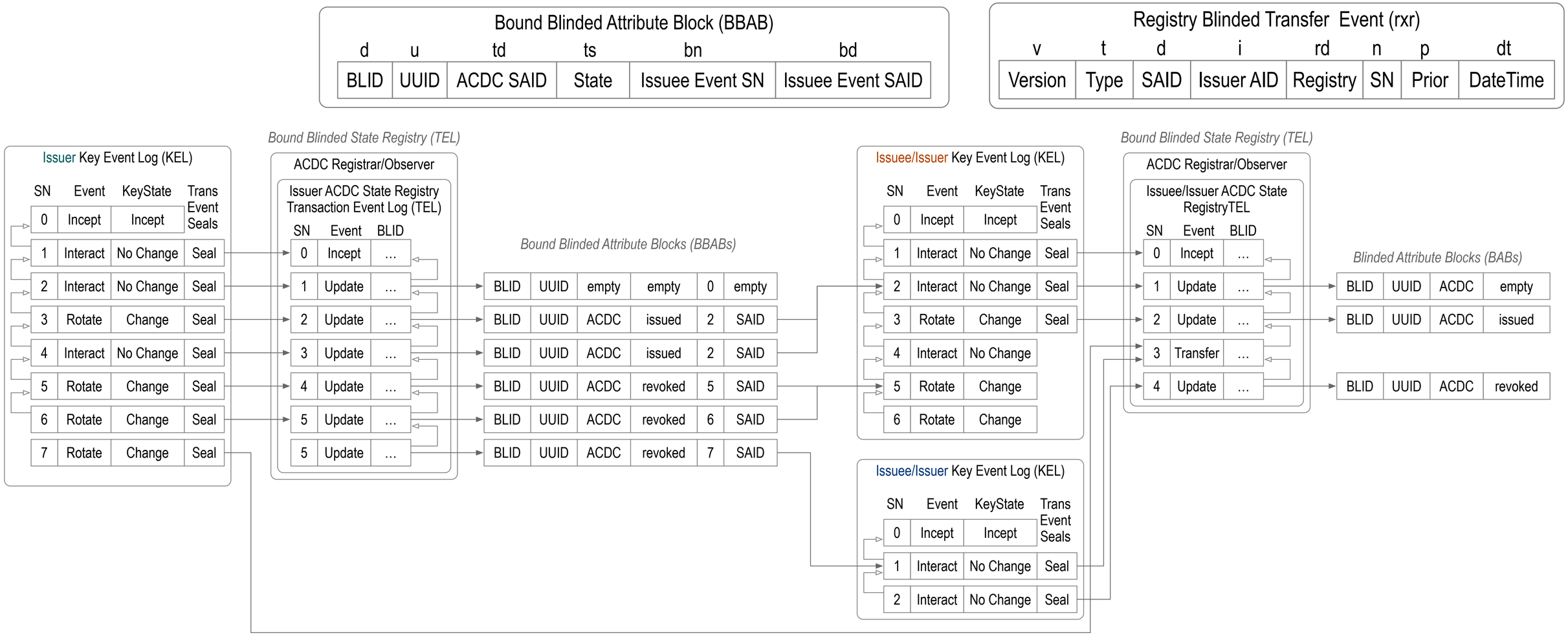
Reverting authority to delegator when delegatee's authority is revoked

Transfer Registry

Replace the controller of registry without reverting the authority to delegator

Key rotation can already transfer control of AID and hence registry to new controllers

New registry transfer *xfr* event to transfer control of registry only.



Safety

The Internet is a dangerous place

Digital Refuge
Online Shelter
Internet Sanctuary
Surveillance Asylum

Safety requires accountability

Privacy requires accountability

Second parties to repeated interactions can correlate the first parties

Any interaction system that does not enable first parties to hold second parties accountable is fundamentally incompatible with privacy

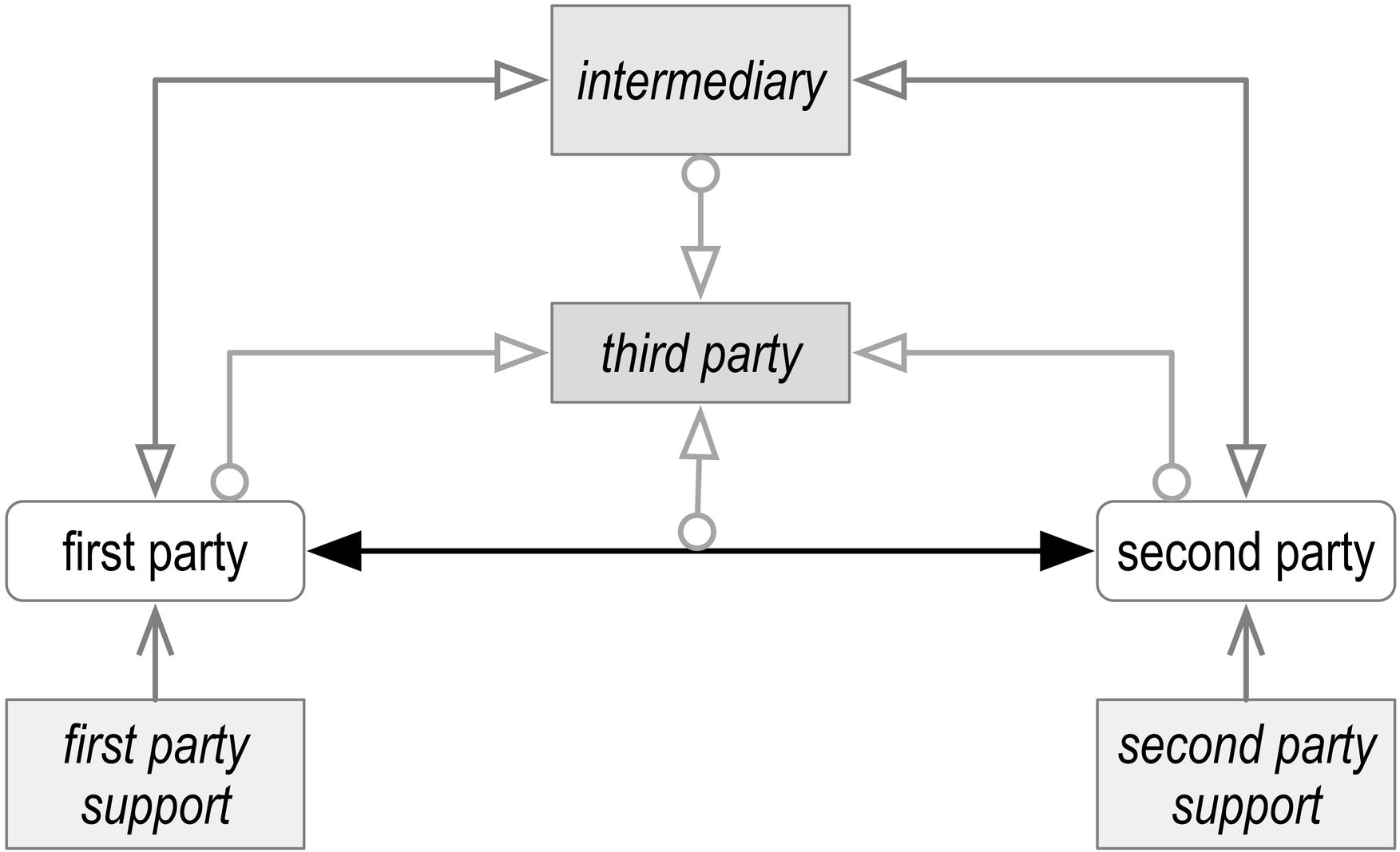
How to enforce accountability?

Control Confidential Contexts

KERI Suite & SPAC/TSP

Create and control safe relationships
where interactions are protected from
surveillance by third parties
and second parties are held accountable for
any information shared with or by them

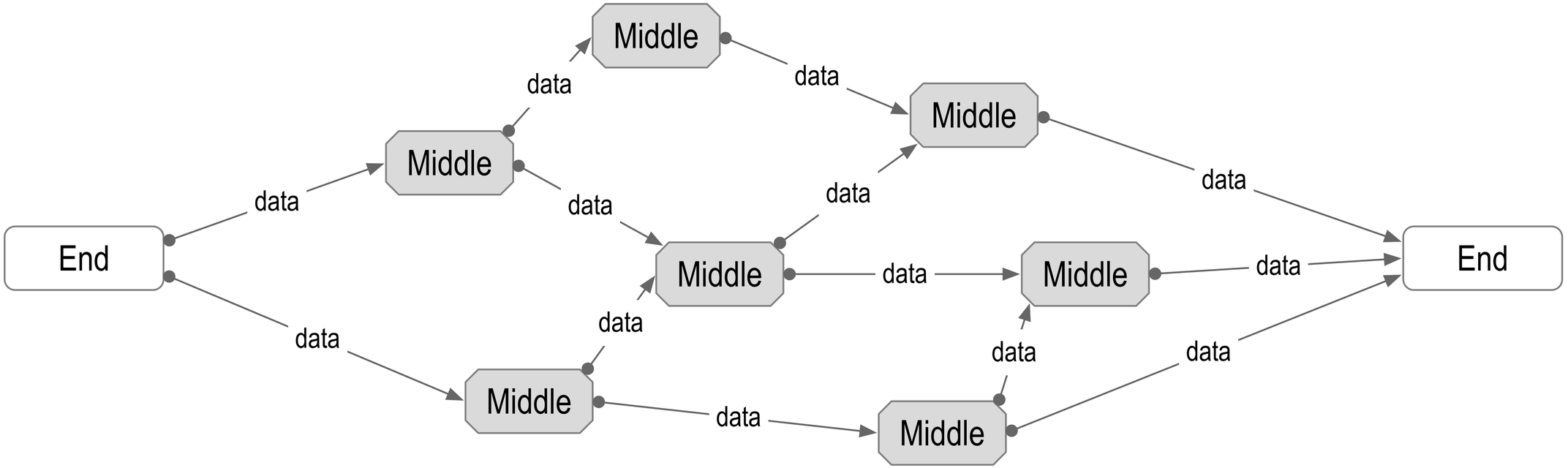
Three Party Information Sharing and Leakage Model



Dual End-Verifiability and End-Only-Viewability

End-to-End Verifiability *of* Authenticity

Only-at-End Viewability via Confidentiality



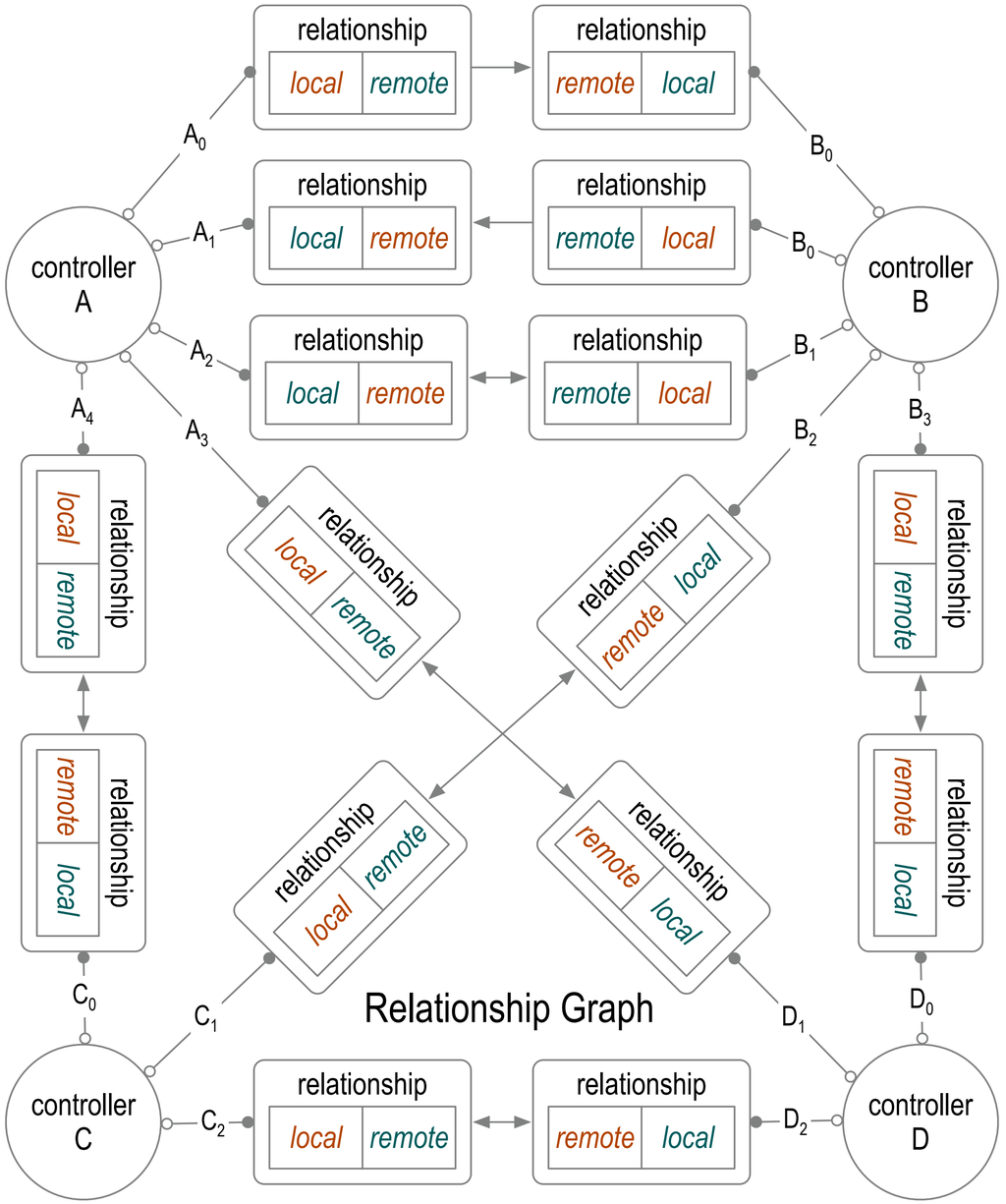
Ambient Verifiability: any-data, any-where, any-time by any-body

End-only Viewability: one-data, one-where, one-time by one-body

Nonrepudiable *Undeniable*

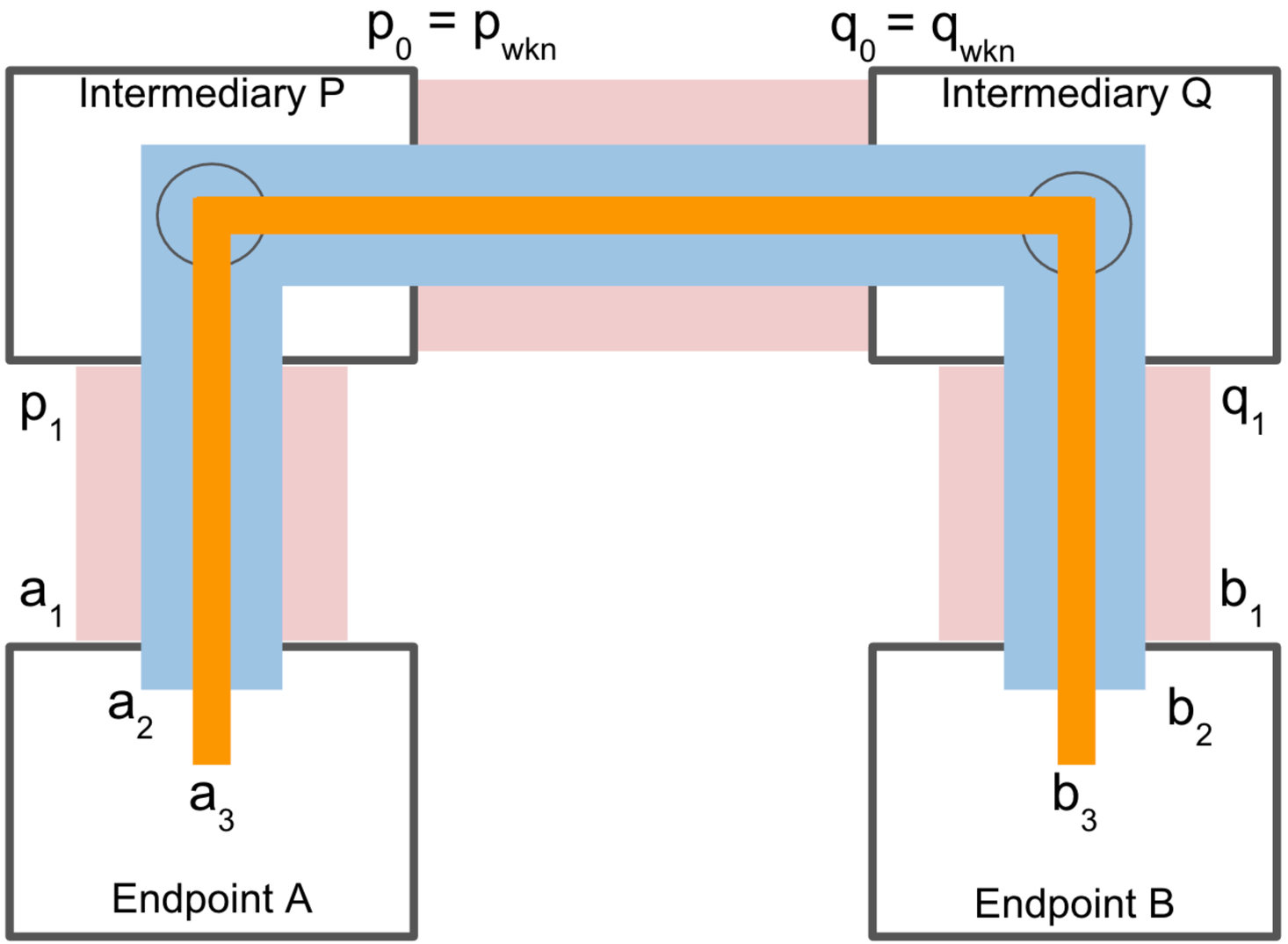
Each person decides with whom they interact, what they share, and why

Relationship Graph (global)



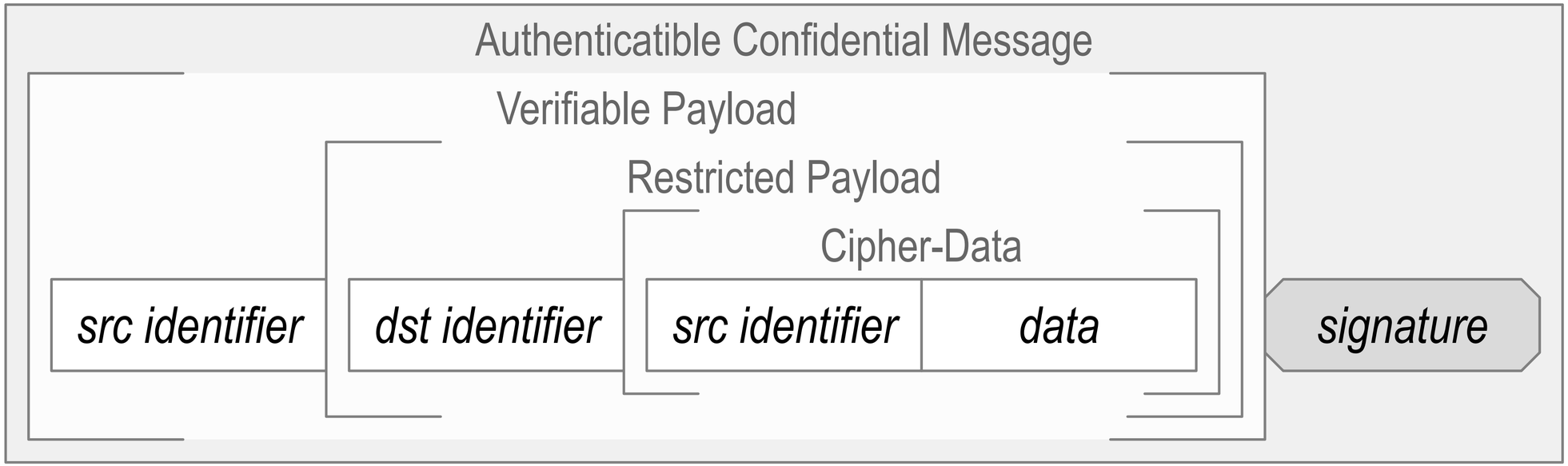
TSP (SPAC) Protocol

Triple-nested tunneling protocol that minimizes surveillable metadata



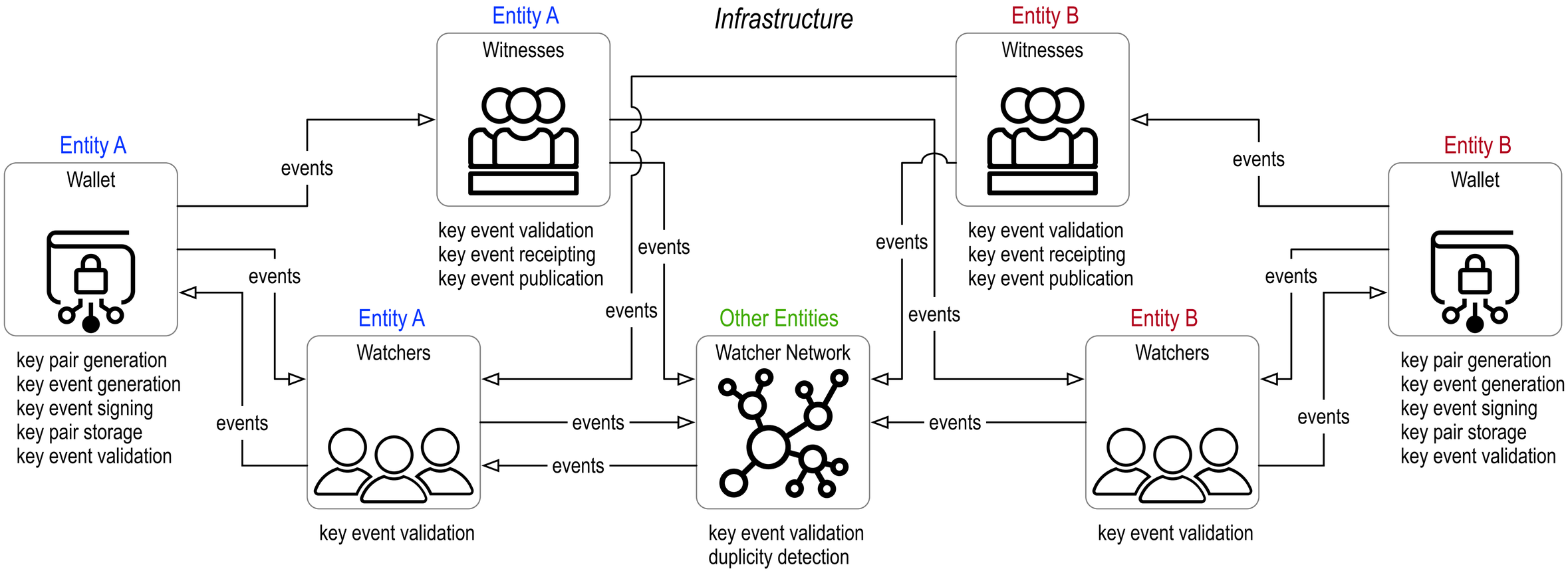
covert vs. clandestine surviellability

Baseline ESSR Message



The Five Ws — One-Click KERI

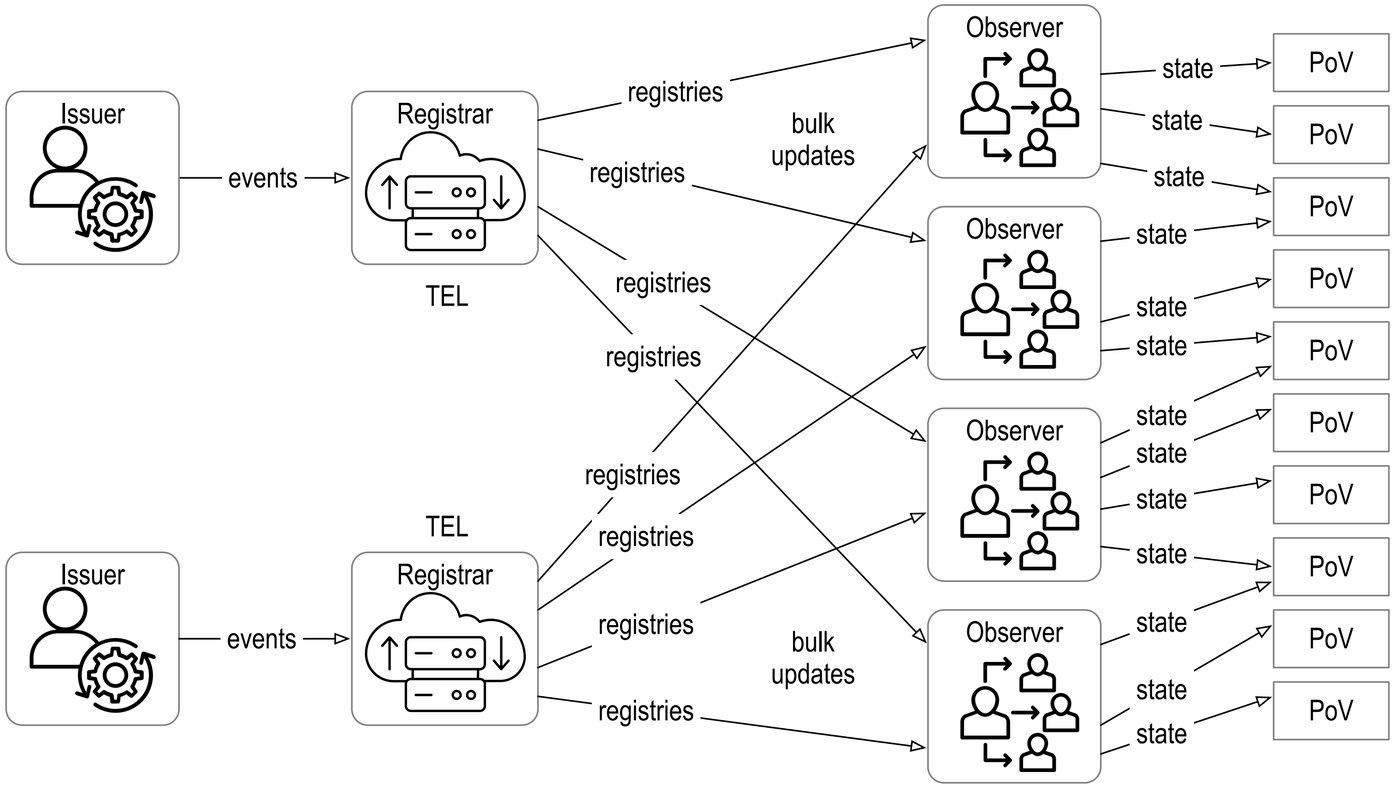
Wallet Witness Watcher Web Wizard



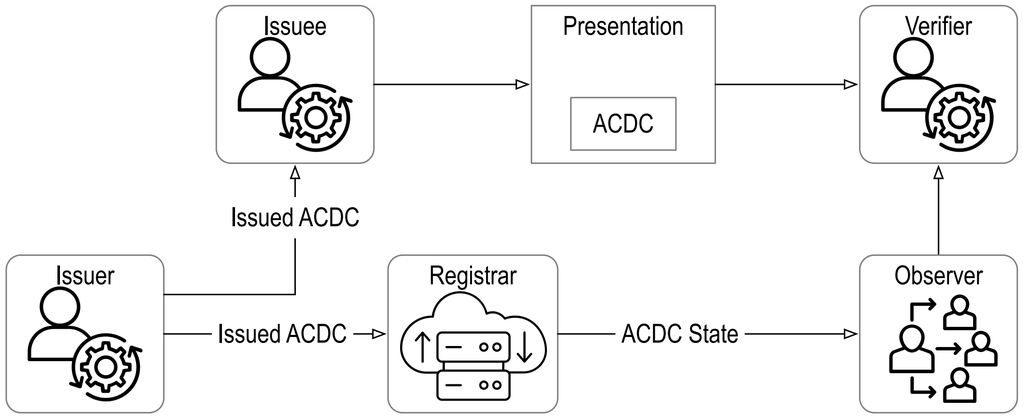
The Five Ws — One-Click KERI

Registrar Observer

SEDI Bulk Issuance

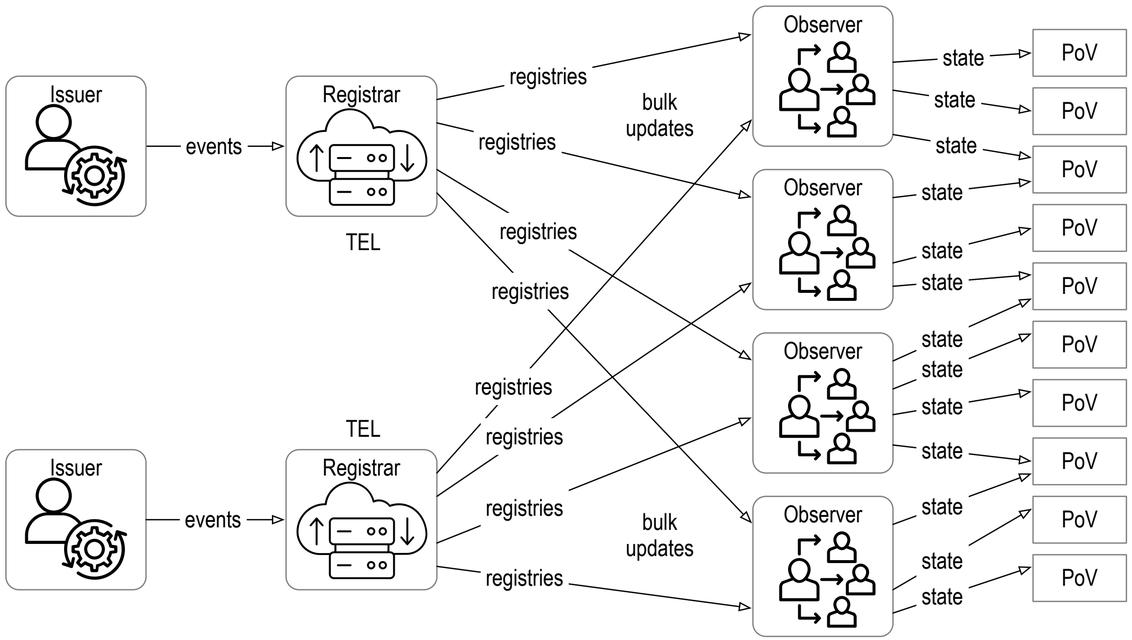


Issuance Verification at Presentation

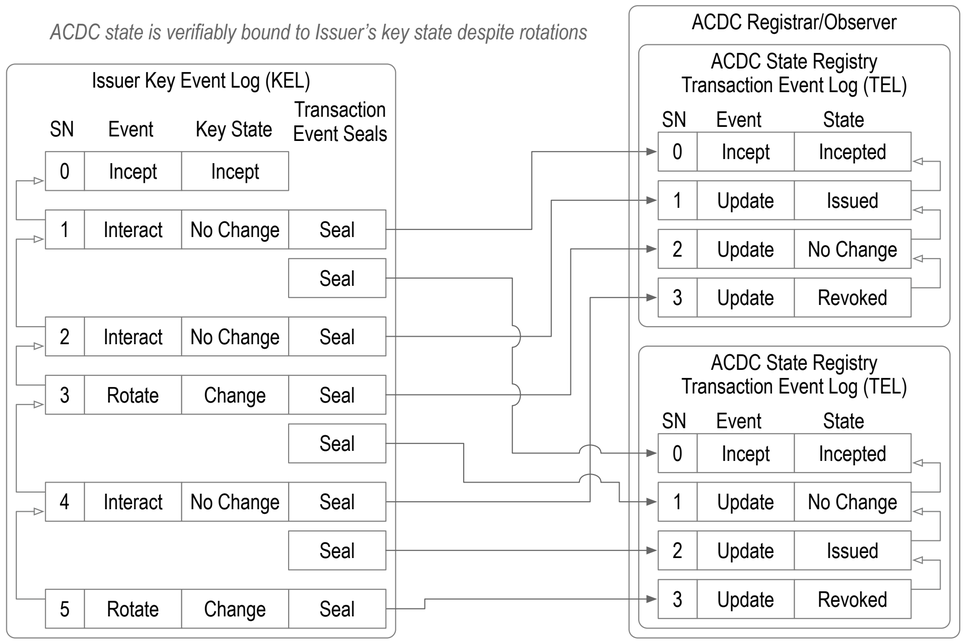


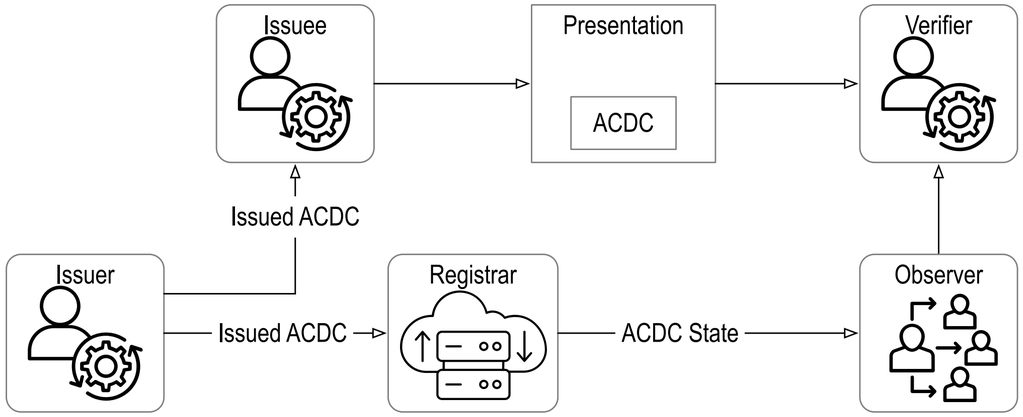
Sparse Merkle Trees

Issuance Registry - ACDC State



ACDC state is verifiably bound to Issuer's key state despite rotations





Blindable Registry = Perfect Forward Privacy

ACDC

v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Registry Inception Event (rip)

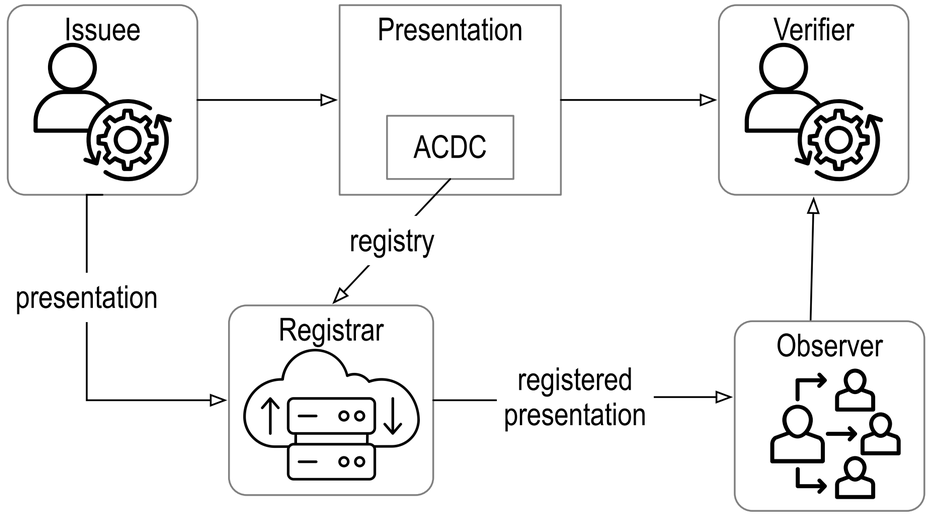
v	t	d	u	i	n	dt
Version	Type	SAID	UUID	Issuer AID	SN	DateTime

Registry Blinded Update Event (bup)

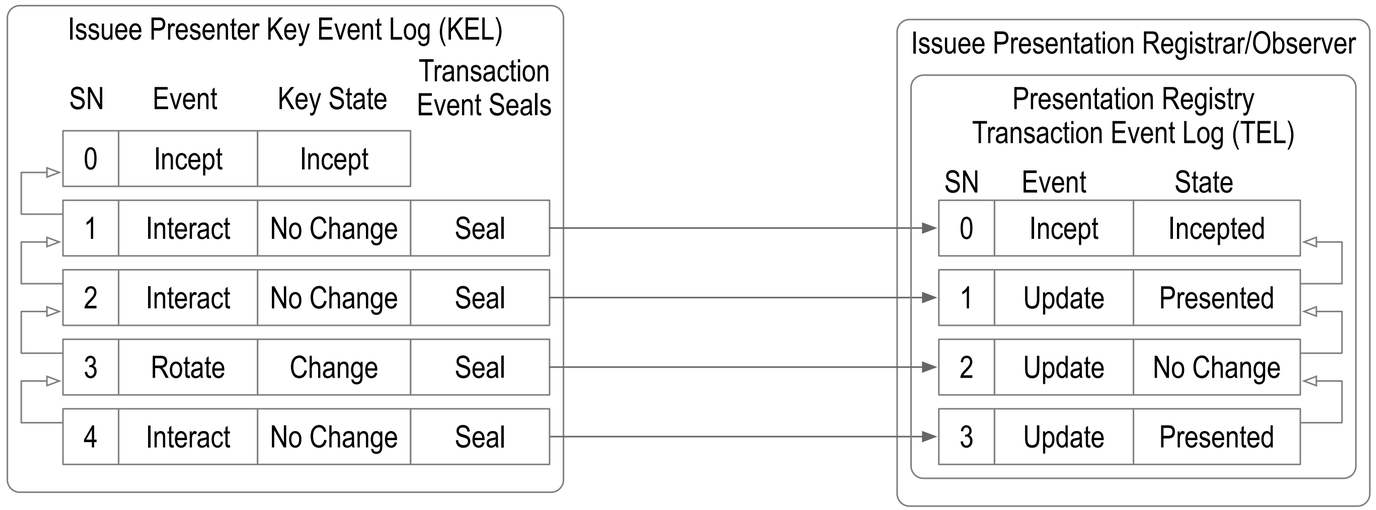
v	t	d	rd	n	p	dt	b
Version	Type	SAID	Registry	SN	Prior	DateTime	BLID

User Fraud Presention Detection

User Presentation Registries



ACDC Issuee Presentation state is verifiably bound to Issuee's key state despite rotations



Identity Revolution

Reputable
Autonomic
Pseudonymity

Control
Confidential
Contexts

RAP-C3

Rapsody

KERI
CONF
2020

KERI
FOUNDATION

KERI

Three Party Information Sharing and Leakage Model

Information is Shared between two parties.

Third party observers may obtain information:

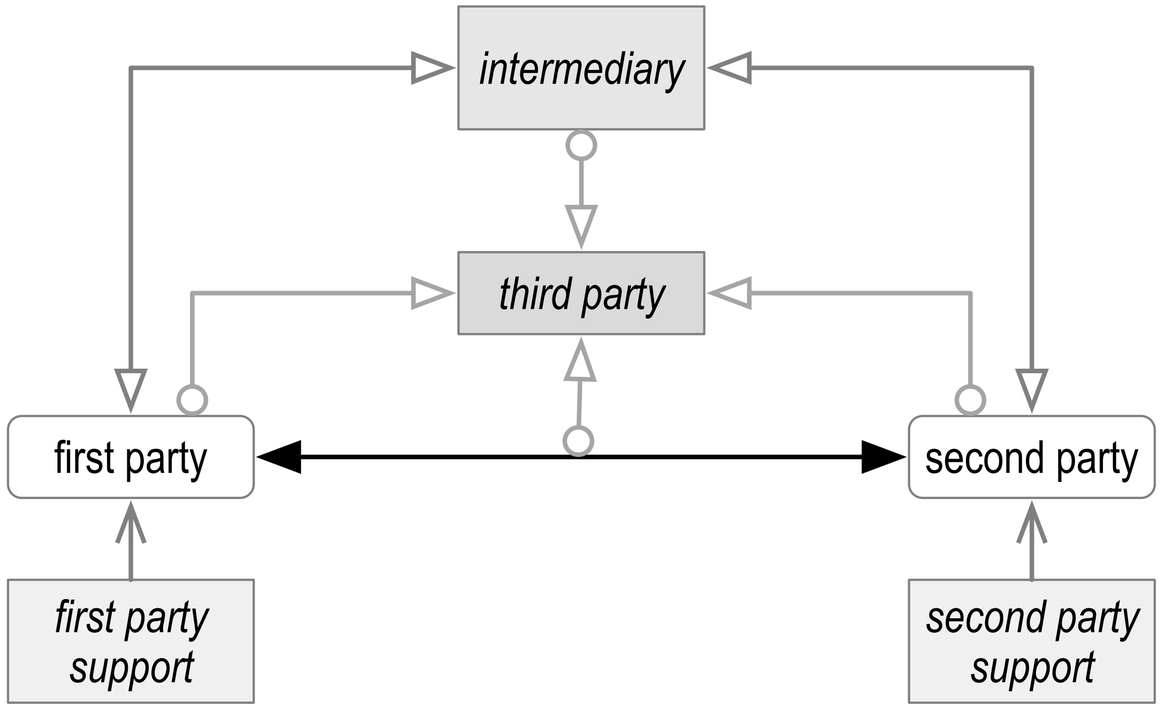
Indirectly as meta-data about the sharing interaction via leakage of the channel.

Directly from either of the first two parties and/or their supporting infrastructure including intermediaries.

Surveillance:

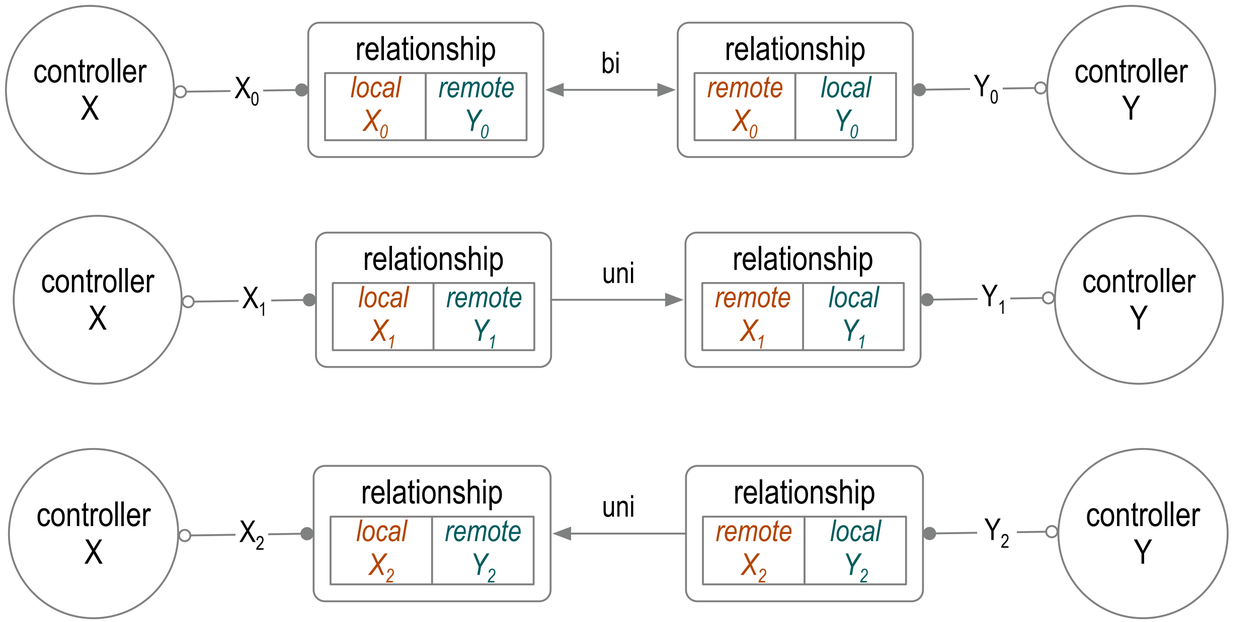
privacy is when third parties have no knowledge of ***who*** is sharing information

confidentiality is when third parties have no knowledge of ***what*** was shared.

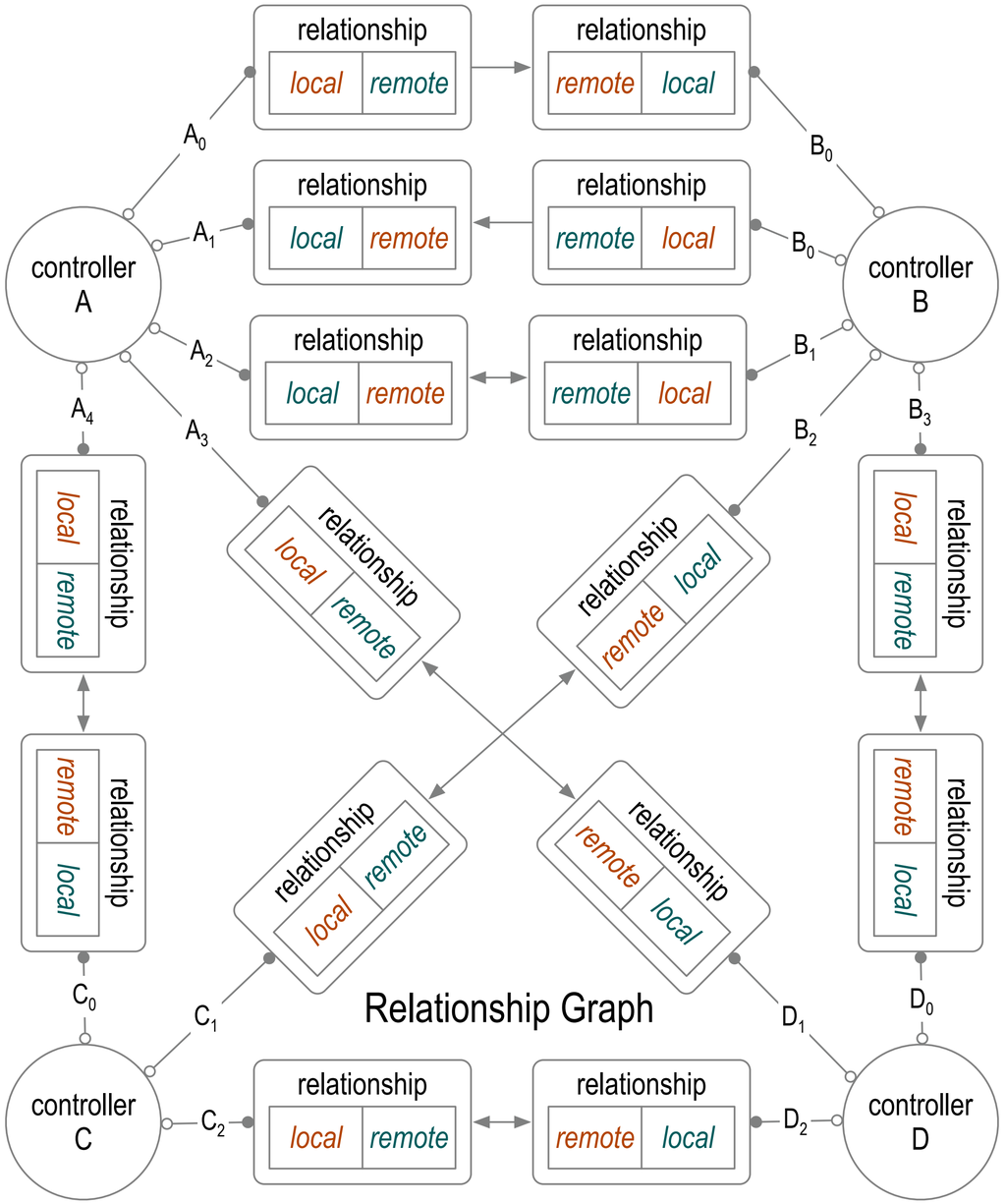


Relationships

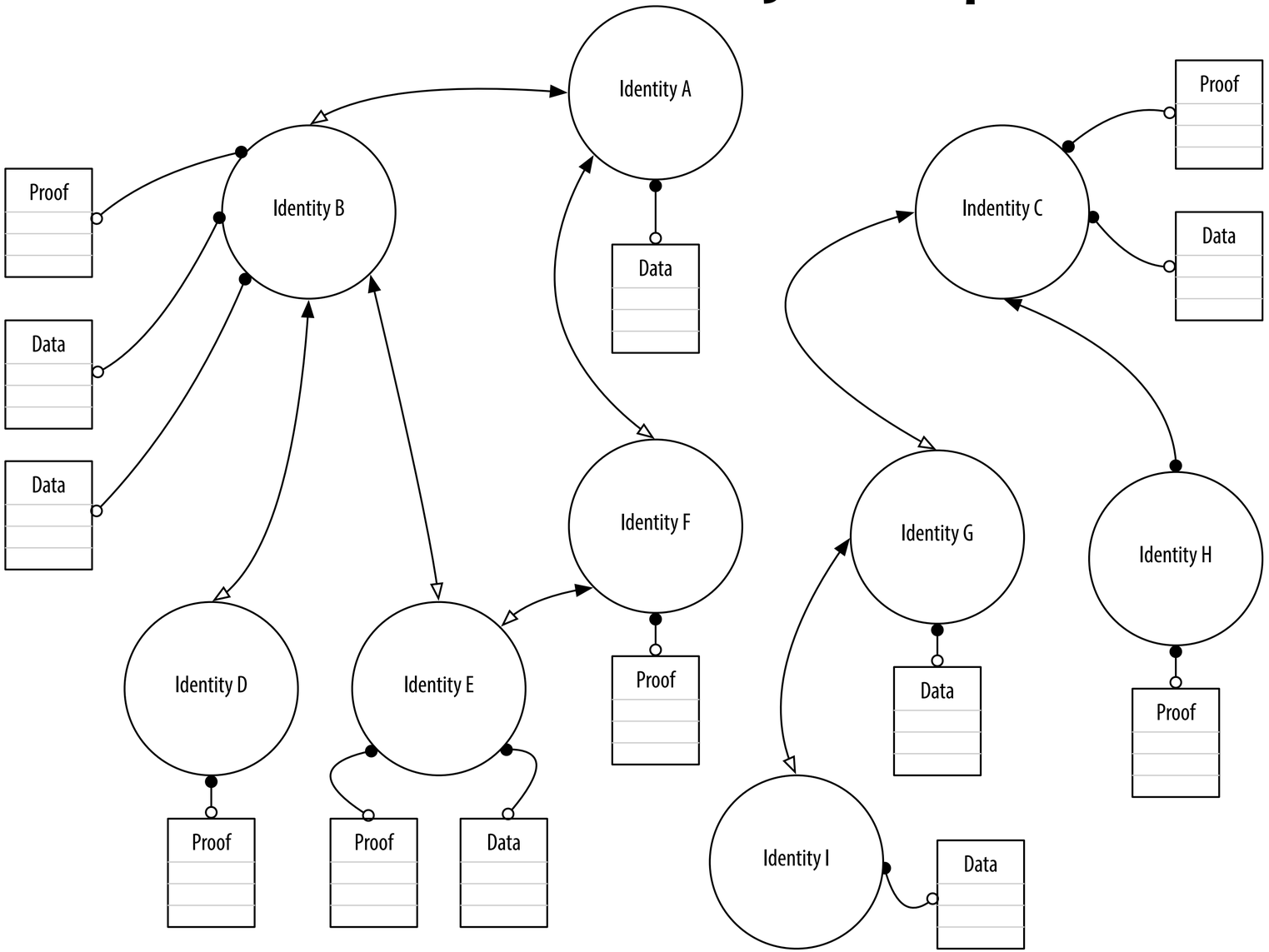
```
(A, <>, B)
(A, ->, B)
(A, <-, B)
```



Relationship Graph (global)



Self-Identity Graph



Relationship graph is the intersection of identity graphs.

Relationship Properties

A ***cryptonym*** is a cryptographically derived pseudonym with at least 128 bits of entropy in the derivation.

An ***AID*** is a cryptonym that is also securely attributable to one or more key pairs.

A ***relationship*** is a pairing of two AIDs, one each from a different *controller*.

*Two different **AIDs** are by themselves uncorrelated in an information-theoretic security sense in that knowledge of one by itself provides no information about the other.*

A ***context*** is a set of events.

*Two **contexts** are disjoint with respect to an AID when that AID appears in one or more events in one set but does not appear in any event in* the *other set.*

A *relationship* is not a *communication channel*, but the events sent over a communication channel may be a ***context*** for the aids in a ***relationship***.

A ***partition*** is a set of *contexts* with mutually disjoint *relationships*.

*Any set of relationships using **ORIs (One Relationship Identifiers)** may form a **partition**.*

Any two-member *contexts* of a ***partition*** may be correlatable due to other information associated with those contexts, but the partitioned relationships by themselves provide no correlatable information, i.e., ***partitioned relationships*** are by themselves not mutually correlatable

An *AID* common to any subset of events within a context provides a *perfectly correlatable* feature across those common events in that context.

Within a context, the *secure attributability* of an AID, together with its *perfect correlatability across that context*, ensures *reputational trust* (good or bad) in that AID.

Partitions balance the concerns of:

strong authenticity and strong confidentiality within a context with,

sufficiently strong privacy between contexts.

PARCC: Portable Authenticatable Reputable Contextual Cryptonymity

ESSR (Encrypt Sender Sign Receiver)

An approach that protects against both KCI and sender impersonation of the ciphertext is called ESSR.

https://eprint.iacr.org/2001/079)

https://neilmadden.blog/2018/11/14/public-key-authenticated-encryption-and-why-you-want-it-part-i/

https://neilmadden.blog/2018/11/26/public-key-authenticated-encryption-and-why-you-want-it-part-ii/

https://neilmadden.blog/2018/12/14/public-key-authenticated-encryption-and-why-you-want-it-part-iii/

The properties that protect against key compromise impersonation attacks are called

TUF-PTXT (Third-party UnForgeability of PlainText),

TUF-CTXT (Third-party UnForgeability of CipherText),

RUF-PTXT (Receiver UnForgeability of PlainText),

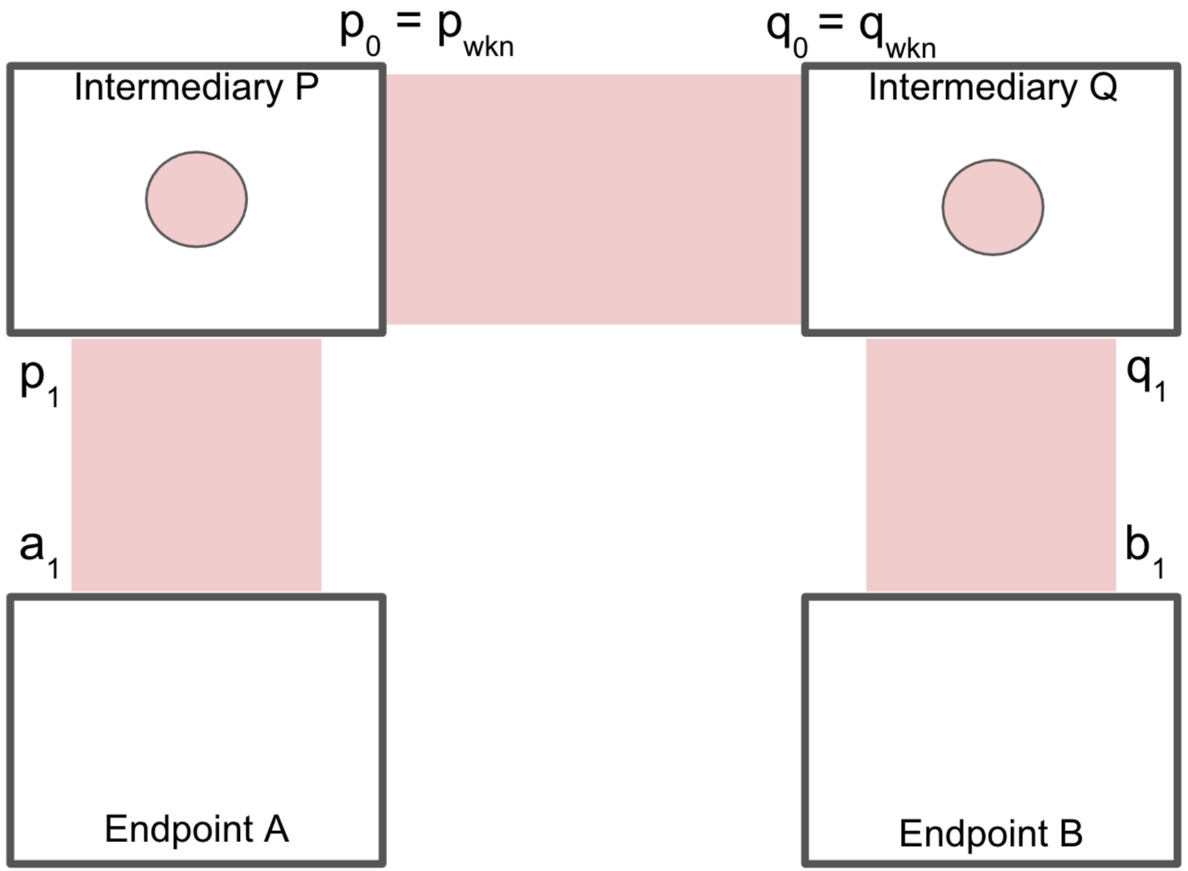
RUF-CTXT (Receiver UnForgeability of CipherText).

Simply using the encrypt then sign approach provides TUF-PTXT, TUF-CTXT, and RUF-CTXT but not RUF-PTXT.

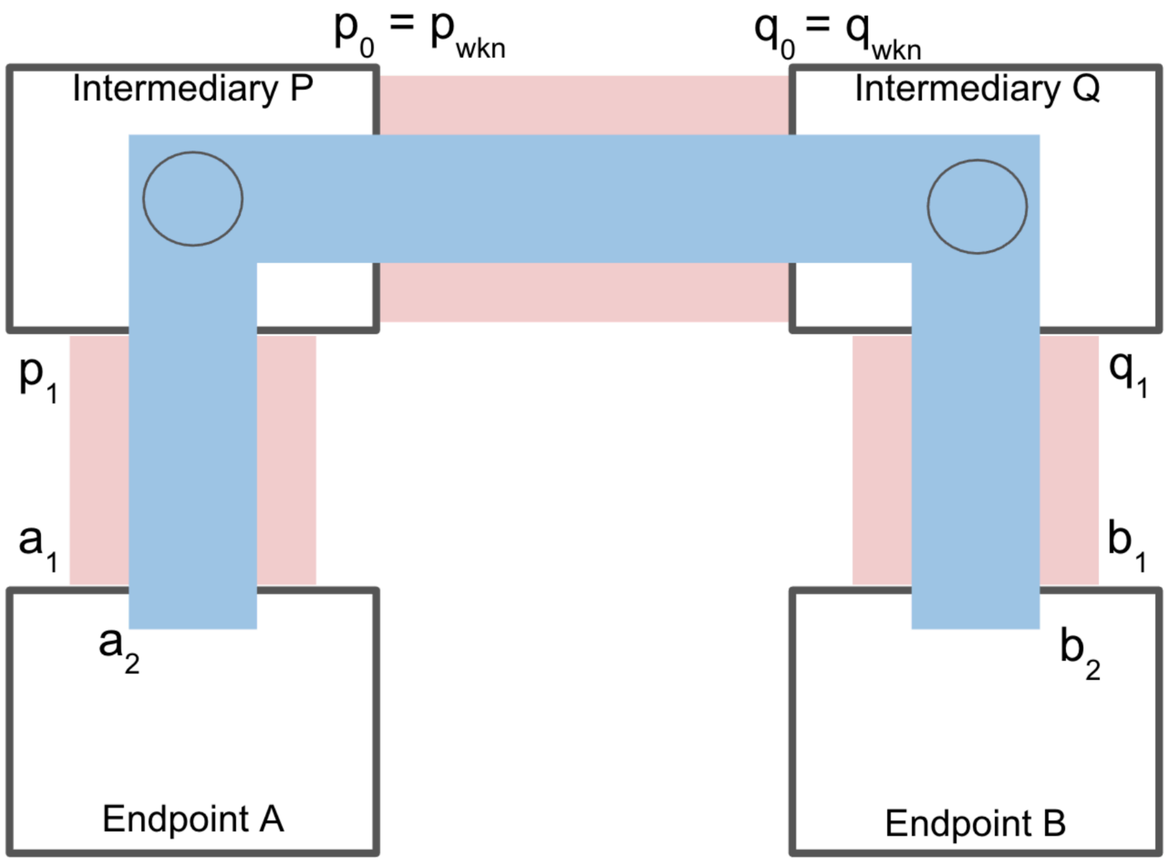
scheme	TUF-PTXT	TUF-CTXT	RUF-PTXT	RUF-CTXT
Encrypt-then-Sign	X	X		X
ESSR	X	X	X	X

As a result of binding the sender's public key inside the ciphertext and binding the receiver's public key in the enclosing signed plain text an adversary is prevented from forging messages that compromise either authenticity or confidentiality. Thus ESSR provides both strong authenticity and strong confidentiality.

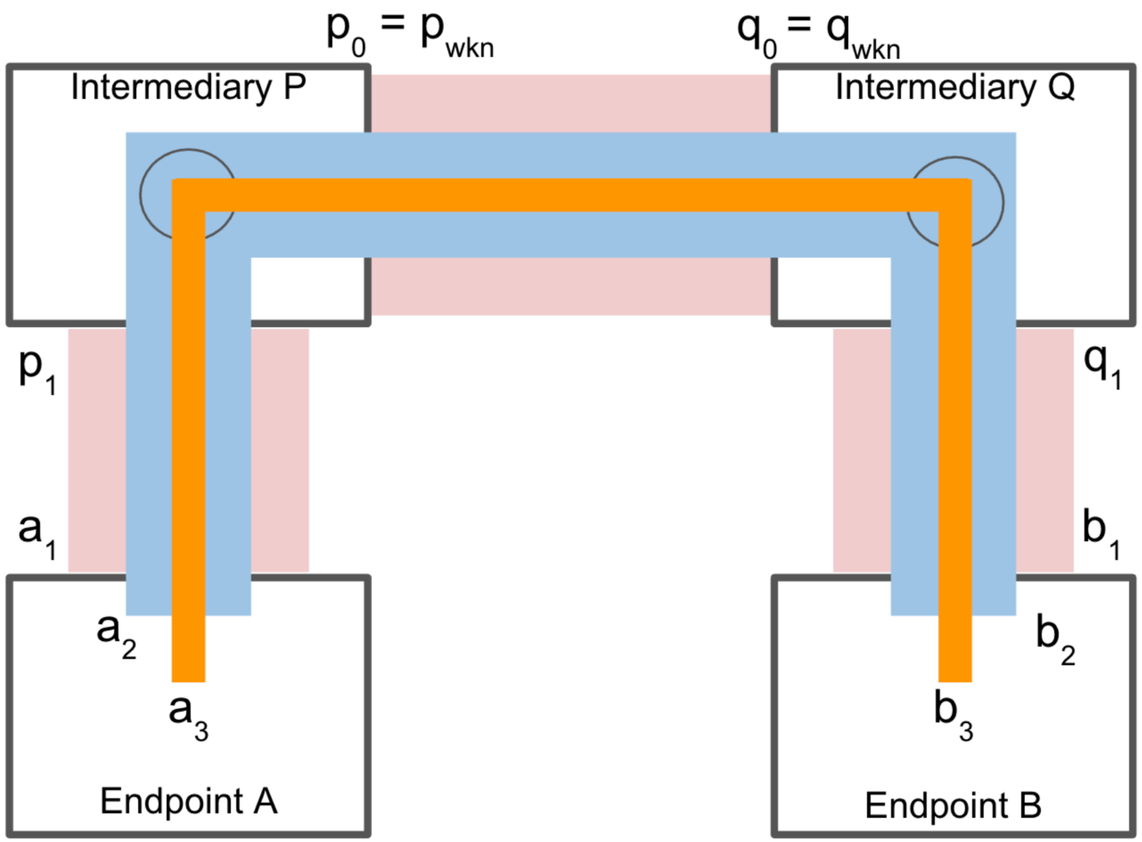
Single Tunnel



Double Tunnel

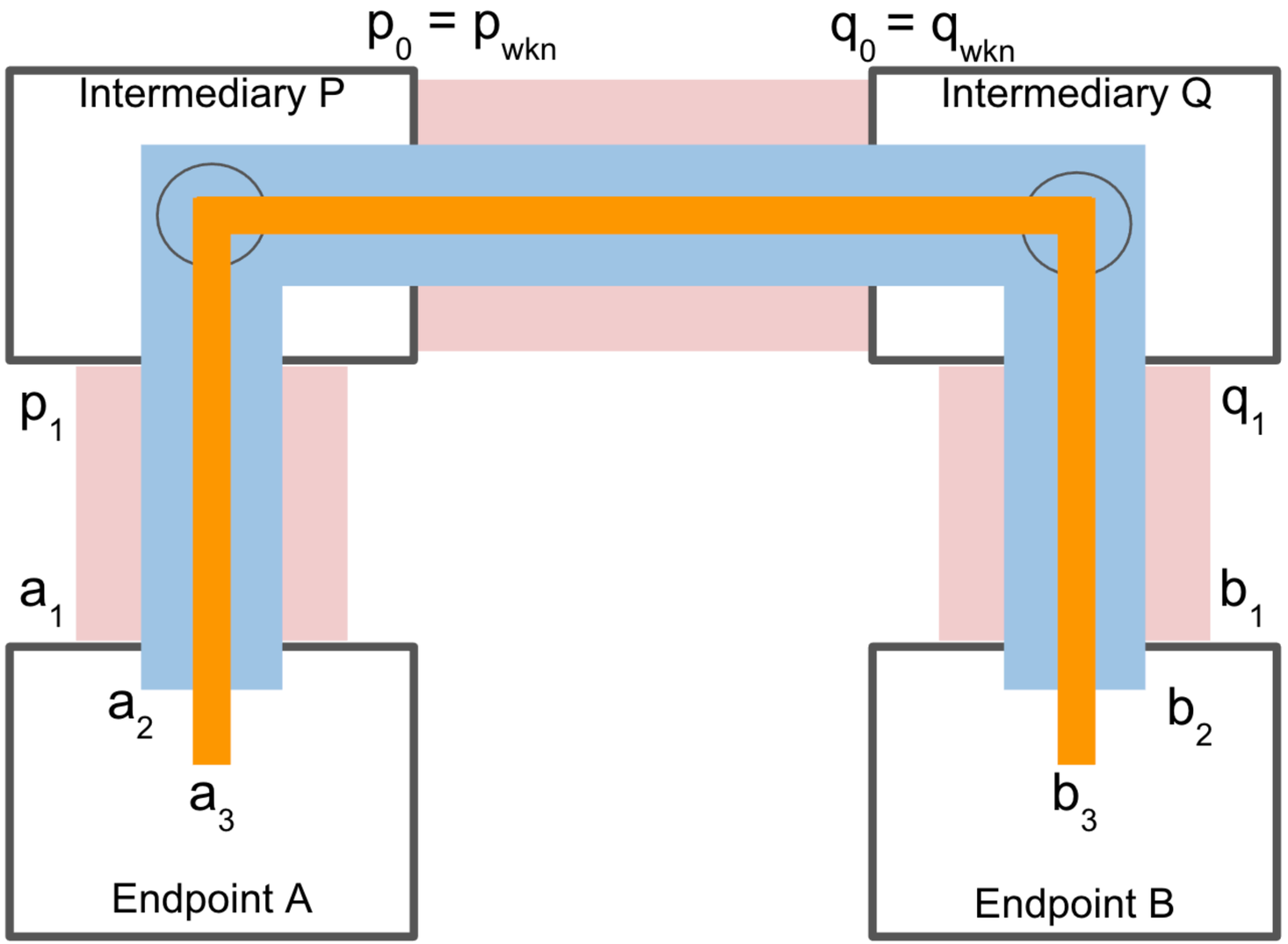


Triple Tunnel



TSP (SPAC) Protocol

Triple-nested tunneling protocol that minimizes surveillable metadata



covert vs. clandestine surviellability

Protocol Syntax

AIDs:

X_0, X_1, X_2

Relationships:

$(X_0, <>, Y_0)$

$(X_1, ->, Z_0)$

$(X_2, <-, Y_1)$

Signatures:

$<data>\ X_0$

$<data>\ (X_0, X_1)$

CipherText:

$\{data\}Y_2$

Messages:

$[src\ X_0, dst\ Y_0, data]$

$<[src\ X_0, dst\ Y_0, \{src\ X_0, data\}Y_0]>X_0$

Interaction Context ID:

$i^0X_0Y_0$

Header:

(version, protocol type, packet type, interaction ID, sequence number, ….)

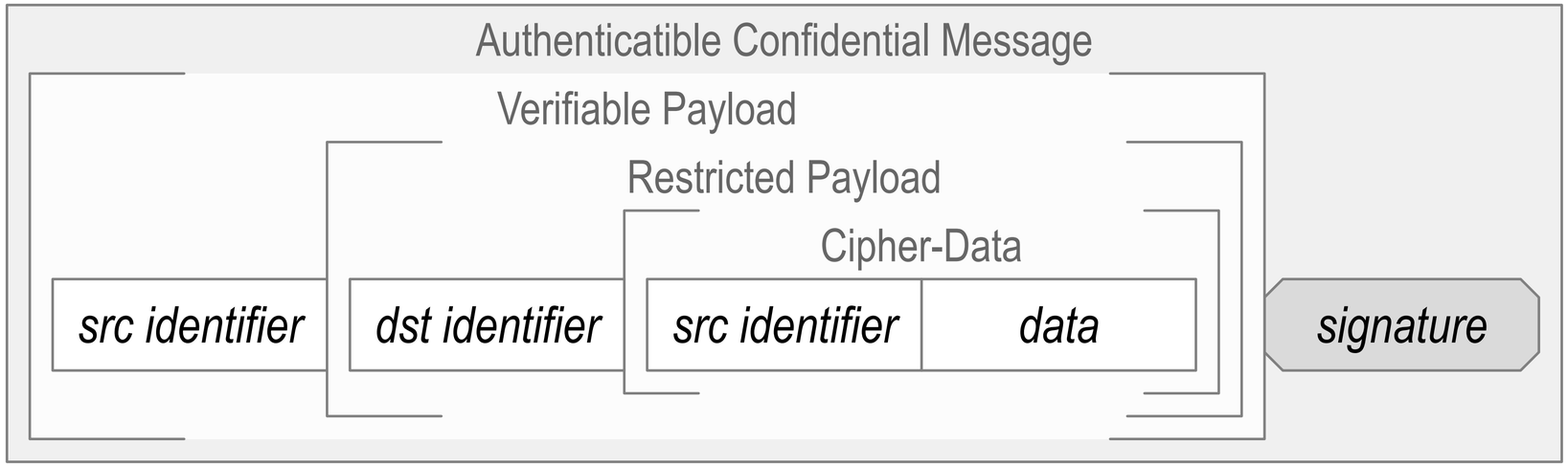
$<[header, src\ X_0, dst\ Y_0, \{src\ X_0, data\}Y_0]>X_0$

Transport (IP) Header: (IP addresses are public by nature with inherently correlatable global context)

$|src\ ip\ X_0, dst\ ip\ Y_0|$

Example:

$|src\ ip\ X_0, dst\ ip\ Y_0|<[header, src\ X_0, dst\ ip\ Y_0, dst\ Y_0, \{src\ X_0, data\}Y_0]>X_0$



Relationship Formation Sub-Protocol

The relationship formation protocol defined above would allow any pre-existing relationship to bootstrap a new relationship without having to perform yet another OOB setup to exchange identifiers. This could also be used to establish new communications context relationships by adding the exchange of the service endpoint IP addresses in addition to the AIDs for the newly formed relationship. This is all accomplished using a no-compromise secure authentic, confidential, and private exchange between the two parties.

Thus, any given relationship that has been sufficiently authenticated and is therefore reputable (not cheap) can be used to issue the equivalent of a letter of introduction for a new relationship, thereby imbuing the new relationship with reputation. We call this an Out-Of-Band-Introduction (OOBI) because the reputable relationship authentication was established Out-Of-Band with respect to the new relationship.

The original communication context relationship does not leak any correlatable information about the newly formed relationship. The only correlatable information is that each communication context relationship AIDs are correlated with the relationship's IP addresses. To reiterate, There is no correlation to any other relationship, including other communication relationships, unless they both use the same or related IP addresses.

The protocol uses interaction identifiers derived from hashes to link together the messages in the formation interaction. Such hash chaining of multi-message protocols where the messages are also signed makes the set of messages in the interaction a verifiable data structure. This is virtually impervious to attack. Moreover, because cryptographic digests are universally unique, using a digest for the interaction ID of a chain of hash-chained digests that form an interaction chain enables a given interaction to span different communication contexts and still be easy to lookup in an interaction database that is indexed by such digests. Finally, this approach makes it easy to detect replay attacks.

ACDC Bound Blinded Registry Example

Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


Registry Placeholder Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "1",
    "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "dt": "2025-08-01T18:06:10.988921+00:00",
    "b": "EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c"
}
```


Virtual Label	Value	Description
d	EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c	BLID (Blinding SAID)
u	aJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa	UUID salty nonce blinding factor, HD generated
td	1AAP	Transaction ACDC SAID field value, top-level d
ts	1AAP	Transaction state value string
bn	MAAA	Bound Issuee key event sequence number field
bd	1AAP	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAaEFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0caJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa1AAP1AAPMAAA1AAP
```


Placeholder updates provide herd privacy wrt correlation to date time of actual state changes

ACDC Bound Blinded Registry Example …

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "2",
    "p": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "dt": "2020-08-02T12:00:20.000000+00:00",
    "b": "EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4"
}
```


Virtual Label	Value	Description
d	EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4	BLID (Blinding SAID)
u	aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTkO	UUID salty nonce blinding factor, HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	0Missued	Transaction state value string
bn	MAAB	Bound Issuee key event sequence number field
bd	EJOnAKXGaSyJ_43kit0V806NNeGWS07lfjybB1UcfWsv	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTkOEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M0MissuedMAABEJOnAKXGaSyJ_43kit0V806NNeGWS07lfjybB1UcfWsv
```

ACDC Bound Blinded Registry Example …

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EDXGqM-V_sfj65Dk-lgMnpOZ9DYziqMaYkrju7NRKpFn",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv"
}
```


Virtual Label	Value	Description
d	EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv	BLID (Blinding SAID)
u	aFudXE-d0b2owzZNBjd78sx4kCTJx-RTP_Zd19HRUcVD	UUID salty nonce blinding factor, HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	Yrevoked	Transaction state value string
bn	MAAI	Bound Issuee key event sequence number field
bd	EDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGvaFudXE-d0b2owzZNBjd78sx4kCTJx-
RTP_Zd19HRUcVDEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4MYrevokedMAAIEDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk
```

ACDC Blinded Registry Example

Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


Registry Placeholder Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "1",
  "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "dt": "2025-08-01T18:06:10.988921+00:00",
  "b": "ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J"
}
```


Virtual Label	Value	Description
`d`	`ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J`	BLID (Blinding SAID)
`u`	`aG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv8`	UUID salty nonce blinding factor, random or HD generated
`td`	`1AAP`	Transaction ACDC SAID field value, top-level `d`
`ts`	`1AAP`	Transaction state value string

Unblinded attribute block as attachment to disclosure of TEL Event


```
-aAYECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6JaG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv81AAP1AAP
```


Placeholder updates provide herd privacy wrt correlation to date time of actual state changes

ACDC Blinded Registry Example ...

Registry Issued Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "2",
  "p": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "dt": "2020-08-02T12:00:20.000000+00:00",
  "b": "EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t"
}
```


Virtual Label	Value	Description
`d`	`EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t`	Dummied BLID (Blinding SAID)
`u`	`aLfCdNAnc-0P2SiruarZSajXiUWu5iU2VfQahvpNCyzB`	UUID salty nonce blinding factor, random or HD generated
`td`	`EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M`	Transaction ACDC SAID field value, top-level `d`
`ts`	`0Missued`	Transaction state value string

ACDC Blinded Registry Example ...

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EM8B1uDhWaJLfpIiEqgp-3EurGUcbfe7u2k5AarDl2XD",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN"
}
```


Virtual Label	Value	Description
d	EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN	Dummied BLID (Blinding SAID)
u	aGx7b16vGHVPT56tX30kYOEzTwiVY4aabc4k9AawYyZG	UUID salty nonce blinding factor HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	Yrevoked	Transaction state value string

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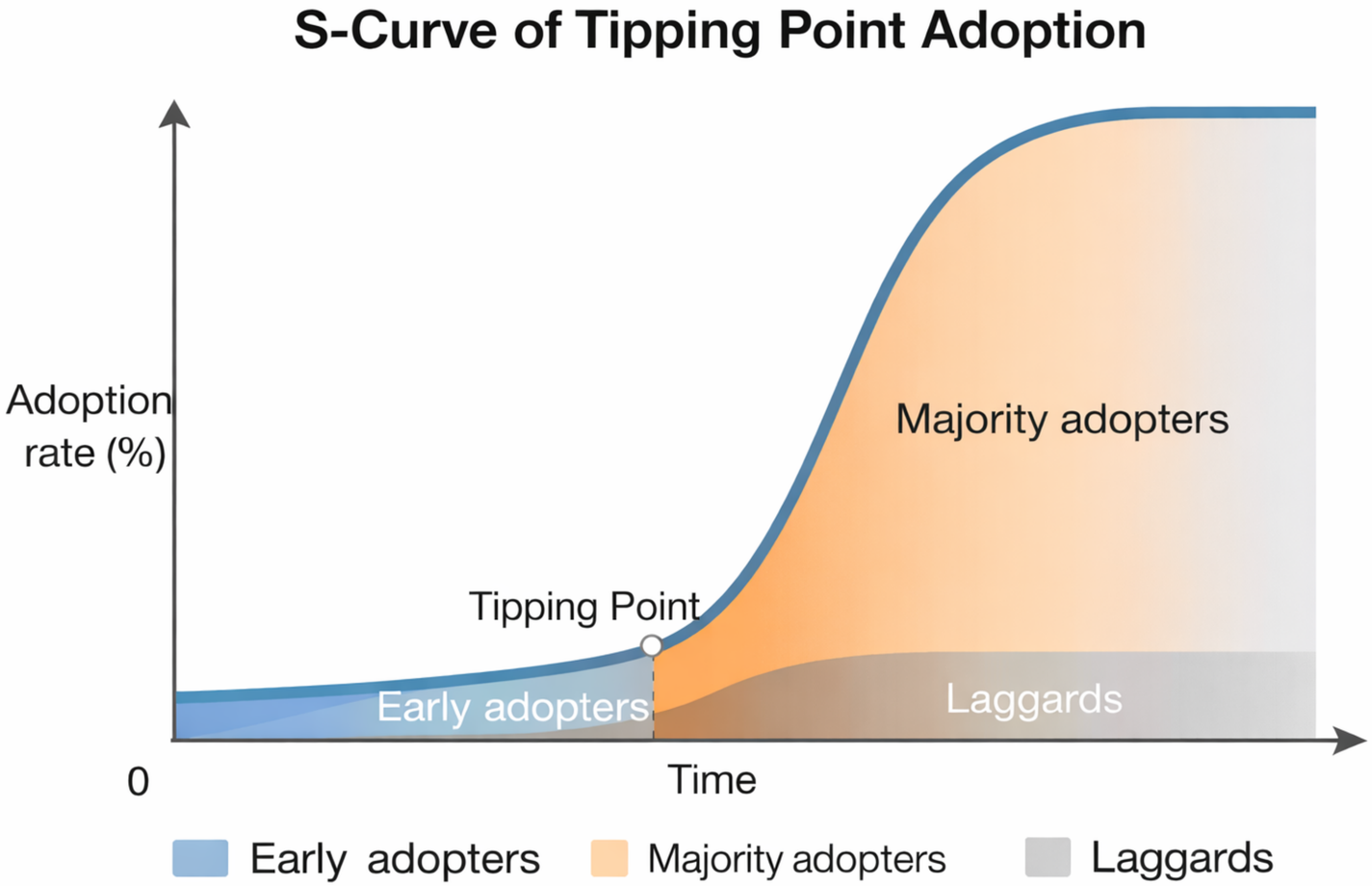
What is KERI *Really*!

Gladwell's Rules for Tipping-Point Adoption

The Law of the Few

The Stickiness Factor

The Power of Context



The Law of the Few

Sufficiently influential set of early adopters Recognize technical suitability

Overcame the 7 stages of adoption resistance: *shock, denial, anger, bargaining, testing, acceptance*

Massive Scale Adoption Vectors

Telecom Sector — GSMA, OVC Alliance

HealthCare Sector — VITAL Initiative, Direct Trust, Carin Alliance, GLEIF

State Entitlements — SEDI

Governor's Office of Economic Opportunity

Widespread Bootstrap Vectors

Organizational Reputational Root-of-Trust — GLEIF vLEI

Personal Reputational Root-of-Trust — Utah State SEDI

All Scale Enablement Vectors

Open Standards Open Source Open Infrastructure ...

The Stickiness Factor

Compelling and contagious

Endemic pathology of fraud → liability/cost incentives *(organizational)*

Regulator imperatives to repress fraud & abuse *(personal & organizational)*

Malaise of digital exploitation → desire to control digital presence *(personal)*

Increasing familiarity with underlying technology *(personal & organizational)*

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The Power of Context Timing and externalities

Specter of AI:

destruction of security, the annihilation of trust, impending doom

- 💣 impersonation → AI impersonation
- 💣 exploitation → AI exploitation
- 💣 surveillance → AI surveillance

Sufficient political will to legislate new civil rights:

Individual control over digital ID, data, & context

Duty of data loyalty

Inevitability of KERI

Severely constrained solution tradespace

Persistent problem cascade that continually exhausts anything less capable

Matter of *when*, not *if*

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Before f 2027, KERI ado reach its ti t!